How Global is Predictability?
The Power of Financial Transfer Learning

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Abstract

We demonstrate that a common global model predicts stock returns more effectively than local models estimated individually for each country, even when the global model is estimated without local data. We introduce a "generalized elastic net" (GENet) to estimate a combined global-and-local model, showing theoretically and empirically that it efficiently (i) transfers information from global data to local countries and (ii) detects unique local components. The resulting model is 96% global—nearly the same function predicts stock returns across countries and over the past century. These findings have broad implications for asset pricing, highlighting the stability of the stochastic discount factor as a function of characteristics.

Keywords: Predictability, asset pricing, transfer learning, machine learning

JEL Codes: C53, C58, F3, G1, G4

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1 Introduction

When researchers and investors seek to determine the expected return on any stock or the stochastic discount factor, should they consider only data on stocks in that country? Or should they consider the predictive patterns of all global stock returns, assuming that the country is irrelevant? Or is global information a useful starting point, but local effects matter too, and, if so, what is the relative importance of local and global information? We answer these questions with "no," "nearly," and "yes, 96% global."

As background, the expected return of any stock i in country c at time t has been found to depend on a range of predictors $x^p_{i,c,t}$. For example, these predictors p include beta, valuation ratios, profitability, and momentum. These predictors are typically studied one at a time, considering how each predicts stock returns, that is, $\mathbb{E}(r_{i,c,t+1}|x^p_{i,c,t}) = \phi^p x^p_{i,c,t}$, focusing on whether ϕ^p is different from zero. Harvey et al. (2016) and Hou et al. (2020) question many of these factors, but Chen and Zimmermann (2022) and Jensen et al. (2023) find a high replication rate.

In any case, optimal portfolio choice and the stochastic discount factor depend on the *joint* multivariate predictive relation, $\mathbb{E}(r_{i,c,t+1}|x_{i,c,t}) = \theta' x_{i,c,t}$, where x contains all the predictors and the vector θ captures each signal's predictive power, *controlling* for the other signals. Asset pricing has long sought a simple model with few characteristics, e.g. Fama and French (1992), but Kozak et al. (2020) show that expected stock returns depend on many characteristics — i.e., there are many non-zero elements of θ.

This asset pricing research was initially done with US data, but global evidence is receiving increasing attention. In particular, looking at each signal one at a time, Jensen et al. (2023) find consistent results across countries in the sense that the univariate predictive coefficients, ϕ^p, tend to have the same signs across countries when estimated separately in each country, broadening the earlier findings of Fama and French (1998, 2012, 2017), Rouwenhorst (1998), Ang et al. (2009) and McLean et al. (2009). In a similar spirit, researchers take note when results do not extend outside the US, for example Fama and French (2012) note that momentum does not appear to work in Japan and Goyal and Wahal (2015) show that a momentum echo fails to extend to a range of countries outside the US.

We ask the deeper question of whether the joint predictive relation that determines the stochastic discount factor is the same in each country. That is, we are not just asking whether each signal predicts returns with the same sign, sign(ϕ^p), but, looking at all signals jointly, whether the overall predictive relation, θ, is the same across countries? In other words, should we use the same function to predict returns everywhere? Or do truly local effects exist? For example, does momentum predict returns with the same strength in each country when controlling for other predictors, and is the relative importance of value and momentum the same in each country? Further, can we exploit global similarities to get better estimates of expected returns in each country? In other words, can we exploit predictive patterns in one area to make better predictions in other areas, an idea called "transfer learning" in the machine learning literature? We address these questions in turn.

Is there a global element to asset pricing? To identify the existence of a global element to asset pricing, we first estimate θ separately in each country using data on stock returns and 111 predictors up to the end of 1999, and then evaluate the out-of-sample (OOS) performance from 2000. For each country, we compare the OOS performance when using the θ estimated based on its own past data with the θ estimated using all global data. We find that the global model performs much better than the local ones, both in terms of OOS R-squared and the risk-adjusted return of the corresponding trading strategy.

To make this finding even more stark, we estimate the global model without using local data. Specifically, we first consider only non-US countries and estimate the global model using only US data. In other words, we "transfer" the US coefficient to the other countries. We find that this transfer model outperforms the local models because (i) the US has much more data, and (ii) apparently this data is helpful outside the US due to a global component.

To show that this effect is not due to anything special about the US, we also perform the analysis in the other way: We estimate θ for, respectively, the US and the collection of all countries outside the US (using all countries to have a lot of data) from 1982 to 2021. We then test the OOS performance on old US data from 1926 to 1981. Again, we find a strong performance of the transfer model, meaning that using a model trained outside the US after 1982 is successful in predicting US stock returns before 1981.

These findings suggest that the function that predicts returns has been relatively stable across countries and over time for the past century! In fact, the estimated θ parameters are 63% correlated across the following five non-overlapping samples: US 1926–1981, US 1982-1999, US 2000+, World-ex-US 1982-1999, and World-ex-US 2000+. This finding shows a remarkable consistency given the noise in estimating the joint predictive power of 111 signals.

Is there a local element to asset pricing? Naturally, the existence of a strong global component to return predictability does not preclude the existence of local effects. As a first look at local effects, we consider whether the local models have any alpha to the global (or transfer) model. We find no alpha, meaning that, based on the global model, also knowing the local models does not add any value. However, this finding could be driven by noise in the local estimates. We can potentially improve the estimate of the local model by relying on the global model without using it directly. This idea can be denoted "soft transfer learning," as opposed to "hard transfer learning" where the global model is used directly without adjustment. For example, suppose that we first estimate the US model. Using this model directly in a country outside the US is hard transfer. Soft transfer, instead, means estimating the deviations from the US model needed for each country.

We develop simple methods to achieve such soft transfer via a two-stage method: (i) First, we estimate the global component via a regression, where parameters are penalized if they deviate from zero (to handle the large number of parameters, as is common in machine learning); (ii) Second, we estimate each country's local component via a regression, where parameters are penalized if they deviate from the global component — herein lies the novel part, which we denote the generalized elastic net (GENet). We show theoretically that the GENet improves forecast precision relative to purely local or purely global models, under certain conditions.

Empirically, we find that our GENet indeed gains power and precision that allows us to identify significant local components. The resulting local-and-global model has a higher R-squared and Sharpe ratio than a purely global model and also outperforms a more naive combination of a purely global and a purely local model.

How global is risk versus mispricing? We study the local components of the predictive model, focusing on the differences between signals that capture risk and those associated with mispricing, relying on the classification of Chen et al. (2022). Interestingly, we find risk-based signals to be purely global, showing no local variation in their predictive relation. In contrast, signals related to mispricing have significant local components in addition to global ones. These findings suggest that local asset pricing arises from local differences in culture, institutional features, or market structure that affect mispricing, but not from risk pricing.

What is the relative importance of local and global pricing? While the local-and-global model outperforms the global model in a statistically significant way, this outperformance is nevertheless economically modest, much smaller than the outperformance of the global model over the purely local model. This finding suggests that the global predictability component is larger than the local components. To quantify this idea, we compute the "relative global component" as the magnitude of the global part of θ divided by the sum of the global and local parts. Using this measure, we estimate that the relative global component is 96%, on average across countries. This finding suggests that predictability is surprisingly similar across countries.

Transfer of non-linear asset pricing. While we have focused so far on linear models, $\mathbb{E}(r_{i,c,t+1}|x_{i,c,t}) = \theta' x_{i,c,t}$, we also show how transfer learning can be used for general non-linear asset pricing models, that is, $\mathbb{E}(r_{i,c,t+1}|x_{i,c,t}) = f(x_{i,c,t})$ for any function f. In particular, we show how to estimate the model, $\hat{f}$, by first using any machine-learning method to capture the global component, and then any other machine-learning method to capture local components. We implement this idea empirically using boosted tree models and neural networks based on random features. We find that these models perform poorly for purely local models, since they are too complex when data is scarce. However, in the context of global transfer learning, these non-linear models perform even better than linear models. In other words, these complex models successfully exploit the rich global data, transferring predictive relations across countries. Moreover, these models are largely global, with small local components, showing a similar degree of global importance as the linear models.

Related literature. The paper is related to several large literatures in addition to the papers already cited. In particular, the international finance literature has theory and evidence on whether assets are priced locally or globally, as reviewed by Karolyi and Stulz (2003). This literature considers whether the CAPM works better in each country with a local market factor or with a global market factor, and, more broadly, whether pricing works better when regressing on a small set of local or global factors. In this regard, Griffin (2002), Hou et al. (2011), Fama and French (2012) and Hollstein (2022) find that pricing errors tend to be lower when stocks are regressed on a local factor model than a global one. Such local pricing can be seen as evidence of some degree of market segmentation.

Whereas this international finance literature regresses stock returns on a few contemporaneous factor returns (i.e., the return of the overall market and various long-short portfolios constructed based on stock characteristics), we regress stock returns directly on ex ante stock characteristics. These differences (predictive regression instead of contemporaneous; with characteristics on the right-hand side instead of factor returns) mean that we can include a full set of characteristics whereas the literature only considers a few factors. We can include more factors for two reasons. First, estimating many factor exposures is difficult, whereas we observe characteristics directly. Second, the factor approach from the international finance literature does not make sense with a general set of factors for the following reason: All stocks can always be perfectly priced by a complete set of factors (or simply the global tangency portfolio), in the absence of arbitrage. Likewise, all local stocks can always be perfectly priced by a complete set of local factors in each country (or just the local tangency portfolio). So considering whether a local or global model has the lowest pricing errors does not make sense when both sets of pricing errors can be made to be zero.

We ask instead whether the function of characteristics that predict returns is the same across countries. Our finding of a strong global component may appear to contradict the finding in international finance that local factor models work best, but the resolution could be that local exposures are better proxies than global exposures for stock characteristics, and expected returns are well captured by stock characteristics. For example, a stock's beta to the local value factor may capture the stock's own valuation ratio better than its beta to a global value factor. In other words, expected returns may be best captured by a nearly global function of stock characteristics. Of course, stock characteristics (or their weighted sum, $\theta' x_{i,c,t}$) can always be seen as the stock's factor loading to some latent factor (i.e., characteristics can always be viewed as exposures, see Lemma 1 of Kelly et al. 2023).

While the international finance approach is well suited to test the CAPM, our approach has other applications. First, our approach is a more powerful way to predict returns, and these expected returns can be used to identify the stochastic discount factor and in asset pricing tests. Second, by including a much more complete set of characteristics, we can determine all of the traits that are relevant for expected returns of global stocks. Third, evidence of local or global return predictability has interesting economic interpretations: A local predictive model can be seen as evidence of differences in risk pricing across countries (rational perspective), cultural effects (behavioral perspective), or differences in institution features (institutional perspective), and our evidence appears consistent with the latter two interpretations.

Further related international finance work includes Rapach et al. (2013) who consider international lead-lad effects, Ilmanen (1995) and Longstaff et al. (2011) who consider local and global effects in sovereign bonds, Brandt et al. (2006) who consider international risk sharing in light of exchange rate movements, and much other work on home bias, flows, integration, and globalization as reviewed by Lewis (2011) and Bekaert et al. (2016).

Our work is also related to the literature on equity predictability using machine learning, see Kelly and Xiu (2023) for an overview. Hard transfer has been found to work for price trends (Jiang et al., 2023), neural networks (Choi et al., 2023), and genetic programming for maximizing the Sharpe ratio (Liu et al., 2021). We complement this literature by presenting a theoretical framework for financial transfer learning, applying this framework across a wide range of factors, isolating local and global effects using novel soft and hard transfer methods, and providing economic interpretations.

The literature on transfer learning is surveyed by Zhuang et al. (2021). The general idea is to use estimates from an area with a lot of data (e.g., recognizing an image of a car) to a related task with less data (e.g., recognizing a truck). Similarly, we start with estimates of predictability from a region with a lot of data and transfer these estimates to predict returns in a country with less data. At a technical level, our method is most closely related to the ridge sequential transfer learning by van Wieringen and Binder (2022), which we generalize to an elastic net, further extend using a transfer parameter controlling how much information is transferred in the spirit of Tommasi and Caputo (2009), and, importantly, we provide a theoretical foundation for such methods, showing how GENet transfer learning enhances predictive power.

In summary, we show that the predictive relation that underlies the stochastic discount factor is remarkably consistent over countries and time. We propose a GENet method to transfer knowledge of the global component across countries and simultaneously extract any local components for each country, demonstrating the method's efficiency gains theoretically and empirically.

2 A Framework for Global and Local Asset Pricing

We consider the excess return, $r_{i,c,t+1}$, of any asset i in country c at time t, where the set of countries is $c = 1, ..., C$, time is discrete $t = 1, 2, ...$, and the set of assets varies across countries $i = 1, ..., N_c$. All excess returns are measured in a common currency—for example, we compute returns in US dollars in the empirical analysis. For each asset, we have a number of predictors collected in the vector $x_{i,c,t} \in \mathbb{R}^p$. For example, $x_{i,c,t}$ contains each asset's valuation ratios, momentum, profitability, and so on.

We are interested in the predictive asset pricing model

$$r_{i,c,t+1} = \theta_c' x_{i,c,t} + \varepsilon_{i,c,t+1} = (\theta_g + \theta_{\ell,c})' x_{i,c,t} + \varepsilon_{i,c,t+1}, \tag{1}$$

where $\theta_c \in \mathbb{R}^p$ is the return-predicting model in country c. The country model consists of two components, $\theta_c = \theta_g + \theta_{\ell,c}$, where $\theta_g \in \mathbb{R}^p$ is a common global component and $\theta_{\ell,c} \in \mathbb{R}^p$ is a local component specific to country c. From a Bayesian perspective, we can think of the data generating process as follows. Nature first draws a global zero-mean parameter, θ_g, with covariance matrix $\sigma_g^2 I$ and independent zero-mean local components, $\theta_{\ell,c}$, with covariance matrix $\sigma_\ell^2 I$, for each country, where I is the identity matrix. Then, returns are generated as the product of the resulting predictive parameter, $\theta_c = \theta_g + \theta_{\ell,c}$, and stock characteristics, $x_{i,c,t}$, plus random noise, $\varepsilon_{i,c,t+1}$ as in (1). This local-and-global model is similar to the hierarchical model of Jensen et al. (2023).

Our analysis considers the following possibilities for the nature of asset pricing:

- **Purely local model.** One possibility is that $\theta_g = 0$ such that the asset pricing model is different in each country. Said differently, the predictive relations in one country are uninformative of the predictive relations in other countries. For example, knowing how momentum predicts returns in the US (controlling for other signals) is not informative about how it predicts returns in Japan.

- **Purely global model.** Another possibility is that $\theta_{\ell,c} = 0$ for all c such that the asset pricing model is the same in each country. This purely global model means, for example, that momentum predicts returns with the exact same coefficient in every country.

- **Global-and-local model.** Lastly, we consider the possibility that asset pricing has both global and local components, that is, $\theta_g \neq 0$ and $\theta_{\ell,c} \neq 0$. The global-and-local model means that predictive relations in one country are informative of predictive relations in other countries, but predictive relations nevertheless also have a uniquely local component. For example, knowing that momentum is a positive predictor of returns in the US is a good starting point in France, but the French predictive relation might nevertheless be weaker or stronger.

These asset pricing models clearly have different implications for investors. The purely local model means that global investors must invest differently in different countries, which likely increases the gains from global diversification. The purely global model means that even a local investor who only trades in domestic stocks should look globally to learn the best strategies. Lastly, the global-and-local model is somewhere in between.

The models also have different implications for asset pricing theory. The global model is consistent with common risk compensations across countries. Specifically, if the global pricing kernel is denoted by m_t and the risk-free rate is r_t^f, we can write (1) as:

$$(\theta_g + \theta_{\ell,c})' x_{i,c,t} = \mathbb{E}_t(r_{i,c,t+1}) = -\mathrm{Cov}_t(r_{i,c,t+1}, m_{t+1}(1 + r_{t+1}^f)). \tag{2}$$

Here, the right-hand side can be seen as the risk exposure, so a global model could be interpreted as saying that the same characteristics correspond to the same risk exposures in each country, under a rational interpretation.[1]

At the same time, the expected returns can be used to identify the pricing kernel. Based on the model for expected returns, the stochastic discount factor is $m_{t+1} = a_t - b_t' r_{t+1}$ with $a_t = \frac{1}{1+r^f_{t+1}} + b_t' \mathbb{E}_t(r_{t+1})$ and

$$b_t = \frac{1}{1 + r^f_{t+1}} \mathrm{Var}_t(r_{t+1})^{-1} \mathbb{E}_t[r_{t+1}] = \frac{1}{1 + r^f_{t+1}} \mathrm{Var}_t(r_{t+1})^{-1} \left[(\theta_g + \theta_{\ell,c})' x_{i,c,t}\right]_{i,c} \tag{3}$$

where the last square bracket is the vector of expected returns based on the global-and-local model.[2] So, if global data helps identify expected returns, it further helps identity the tangency portfolio and the pricing kernel.

Under a behavioral interpretation, a global model can be seen as evidence of common behavioral biases across countries, unaffected by cultural differences. In contrast, any local effects imply that markets operate differently across countries due to differences in risk, behavior, or institutional features.

The asset pricing model (1) is decidedly simple from an economic perspective, but it is not trivial to estimate efficiently given the large number of parameters. Indeed, in our empirical analysis, we consider 111 predictors across 34 countries, yielding nearly 4,000 parameters in total.

In fact, just estimating 111 parameters is a challenge in many countries, especially when the number of stocks is low and the available time period is short. However, having a global component in (1) means that global data is useful even in a local context. Being able to use information about the predictive relation from global data is very helpful, assuming that this relation also works locally.

The problem of having little data in one context, but rich data in a related context, is not unique to equity return prediction, and approaches to tackle this issue are studied in the transfer learning literature. For example, the transfer learning literature estimates a neural

[1] We note that this interpretation does not necessarily imply that markets are integrated. For example, if the same combination of characteristics proxy for the local market beta, then the global model could apply even without market integration.

[2] To see that this m is the (minimum variance) stochastic discount factor (Hansen and Jagannathan, 1991), note that a few calculations show that $E_t(m_{t+1}(1 + r^f_{t+1})) = 1$ and $E_t(m_{t+1} r_{t+1}) = 0$, which are the properties of a stochastic discount factor.

network for one task, and then re-uses this network for another task, just re-estimating the final layer of parameters.

We consider the following simple transfer learning methods that are designed for our local-and-global asset pricing model (1), but should also be highly useful in other contexts. The methods are summarized in Table 1 and we discuss them in turn.

Models	**Estimation methods**		
	No global benchmark	Global benchmark using all global data	Global benchmark using data outside country of interest
Local model	local		
Global model		global	hard
Local-and-global		2stage, joint	soft

Table 1: Overview of Models and Estimation Methods.

1. **Local estimation (local).** Under the assumption of a purely local model, the natural estimation strategy is to estimate each country's predictive relation, $\theta_c = \theta_{\ell,c}$, separately in each country.

2. **Pooled global estimation (global).** Under the assumption of a purely global model, one estimation strategy is to pool all global data and then estimate the common predictive relation, θ_g, using this pooled data.

3. **Hard transfer learning (hard).** If the pooled global estimation works in any country c, then this could be evidence of global asset pricing, but it could also be because the country's own data is used as part of the pooled estimation. So, an even cleaner way to establish a non-zero global effect (a sort of empirical "existence proof") is to predict returns in a country c based on parameters estimated using only foreign data. Using parameters from one task (e.g., predicting US stock returns) to solve another task (e.g., predicting Chinese stock returns) can be called hard transfer learning, where "hard" refers to the parameters being transferred directly without any local modification. For example, we estimate the predictive relation using US data and then apply it in other countries for a later time period (out-of-sample in both geography and time).

4. **Soft transfer learning (soft).** One simple way to use both local and global information is to first make separate local and global estimates (1 and 2 from above) and then use a linear combination of these. This simple method does not work empirically in the sense that it produces predictions that are worse than those from a purely global estimation or a hard transfer model, as we later show. A more powerful way to identify local information is to build off the hard transfer estimates, and then estimate additional local effects, yielding what we refer to as soft transfer learning. Soft transfer learning aims to take advantage of similarities in the conditional return distributions across countries. Relying partly on global data is helpful because the transfer of information reduces noise such that local relations become more visible. Said differently, by transferring part of the signal, the signal-to-noise ratio increases. For example, we can estimate the predictive relation in the US, and then use this relation in Denmark, plus a Denmark-specific component. In a small sample, it may be easier to estimate how Denmark differs from a well-estimated global relation than to estimate the Danish predictive relation from scratch.

5. **Two-stage global-and-local estimation (2stage).** Another approach to estimate a global-and-local model is to first estimate the global component using a pooled estimation and then add local components in a second-stage estimation. Formally, this two-stage procedure seeks to estimate θ_g in the first step and then estimate $\theta_{\ell,c}$ in the second step.[3] This two-stage method is similar to the soft transfer learning method, but the advantage of the two-stage procedure is that the first step is based on more data (all global data) than the soft transfer learning (which only uses US data).

6. **Joint global-and-local estimation (joint).** Rather than estimating the global-and-local in a two-stage transfer-learning procedure, we can also estimate the model directly using a joint estimation of θ_g and $\theta_{\ell,c}$, that is, a simultaneous one-step method. In principle, a one-step procedure combines a full use of all data and a consistent

[3]The soft transfer learning is based on the same idea, but the first step actually estimates $\theta_g + \theta_{\ell,US}$ in the example in which we transfer information from the US to other countries. Hence, the second step of the soft transfer learning methods estimates $\theta_{\ell,c} - \theta_{\ell,US}$ in each country c, thus arriving at the full predictive relation $\theta_g + \theta_{\ell,c}$ (by summing the first and second steps).

treatment of the data generating process. The cost is more computational complexity, so it is an empirical issue whether such a one-step joint method works well.

3 Statistical Methods: Financial Transfer Learning

We next present the statistical tools used in our empirical estimation the local and global models (Sections 3.1–3.3). We also present theoretical results on benefits of our transfer learning methods (Sections 3.4–3.5). Readers most interested in the empirical findings can go directly to Section 5, or just skip the theoretical results in Sections 3.4–3.5.

We begin by expressing the regression equation (1) in vectorized form for each country $c \in \{1, ..., C\}$:

$$Y_c = X_c(\theta_g + \theta_{\ell,c}) + \varepsilon_c, \tag{4}$$

Here, $Y_c \in \mathbb{R}^{N_c}$ is the vector of return observations in country c and $X_c \in \mathbb{R}^{N_c \times p}$ contains the corresponding characteristics (or covariates) for all stocks in the country. Further, $\theta_c = \theta_g + \theta_{\ell,c}$ is country's parameter vector, where θ_g is the global component and $\theta_{\ell,c}$ is the local component specific to country c.

3.1 Single-Region Methods: Elastic Net (ENet)

The simplest methods to estimate are the local, global, and hard transfer learning methods since each of these consists of estimating parameters in one region at a time. Given the large number of characteristics that may predict stocks returns, we estimate each of these models with an elastic net, a popular method in machine learning. For example, applying an elastic net in country c means estimating θ by minimizing the following objective:

$$\hat{\theta}^{\text{ENet}}_{c,\lambda,\alpha} = \underset{\theta \in \mathbb{R}^p}{\arg\min} \{(Y_c - X_c\theta)'(Y_c - X_c\theta) + P_{\lambda,\alpha}(\theta)\}. \tag{5}$$

Here, the first term, $(Y_c - X_c\theta)'(Y_c - X_c\theta)$, is the squared sum of prediction errors. The second term, $P_{\lambda,\alpha}(\theta)$, induces parameter shrinkage by penalizing large parameters:

$$P_{\lambda,\alpha}(\theta) = \lambda(\alpha\|\theta\|_1 + (1-\alpha)\|\theta\|_2^2) \tag{6}$$

with $\lambda \geq 0$, $0 \leq \alpha \leq 1$, $\|\theta\|_1 = \sum_k |\theta_k|$, and $\|\theta\|_2^2 = \sum_k (\theta_k)^2$. We apply the ENet in three different ways.

Local. A local method simply estimates the parameter θ_c for any country c using the ENet (5) based on its own data

$$\hat{\theta}_c^{\text{local}} = \hat{\theta}_{c,\lambda_0,\alpha_0}^{\text{ENet}} \tag{7}$$

Hard transfer. The hard transfer estimator instead uses data from *another* country, $c_0 \neq c$:

$$\hat{\theta}_c^{\text{hard}} = \hat{\theta}_{c_0,\lambda_0,\alpha_0}^{\text{ENet}} \tag{8}$$

Global. Further, the global estimator uses data from all countries, indicated by having a subscript $\mathcal{C} = \{1, ..., C\}$:

$$\hat{\theta}_c^{\text{global}} = \hat{\theta}_{\mathcal{C},\lambda_0,\alpha_0}^{\text{ENet}} \tag{9}$$

In short, these single-region methods can be estimated with a standard penalized regression. It is well-known that this form of penalized regression yields a smaller variance at the cost of having a bias towards zero, compared to ordinary least squares estimation (OLS). A good choice of the penalty function can lead to a considerable decrease in expected prediction errors, especially if the number of parameters, p, is large. The elastic net has well-known special cases: $\lambda = 0$ is a standard OLS regression; $\alpha = 0$ is denoted a ridge regression; and $\alpha = 1$ is called a Lasso regression.

3.2 Two-Step Learning: Our Generalized Elastic Net (GENet)

We next consider how to use the fact that the model (4) has both local and global elements. Suppose that, before estimating the parameter θ for country c, we have an initial guess $\hat{\theta}_0 \in \mathbb{R}^p$ (e.g., from another region). We then define the generalized elastic net (GENet) estimator based on this guess:

$$\hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,\alpha}(\hat{\theta}_0) = \underset{\theta \in \mathbb{R}^p}{\arg\min} \{(Y_c - X_c\theta)'(Y_c - X_c\theta) + P_{\lambda,\alpha}(\theta - \nu\hat{\theta}_0)\}. \tag{10}$$

The GENet minimizes the sum of the squared prediction errors plus a penalty $P_{\lambda,\alpha}(\theta - \nu\hat{\theta}_0)$, where P is defined in (6). Intuitively, this method balances two objectives: fitting the local data well and shrinking the solution toward $\nu\hat{\theta}_0$. When the initial guess is zero, GENet reduces to the standard elastic net from (5), that is, $\hat{\theta}^{\text{GENet}}(0) = \hat{\theta}^{\text{ENet}}$.

The idea of transfer learning is that the initial "guess" $\hat{\theta}_0$ is actually estimated using another region. We propose two stragegies: "soft transfer" and "two-stage."

Soft transfer. We first obtain a local estimate from another country $c_0 \neq c$, denoted $\hat{\theta}^{\text{hard}}_c$ in (8), and use it as input to GENet:

$$\hat{\theta}^{\text{soft}}_c = \hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,\alpha}(\hat{\theta}^{\text{hard}}_c) \tag{11}$$

Two-stage. We first estimate a global parameter $\hat{\theta}^{\text{global}}_c$ from (9), and use it as input to GENet:

$$\hat{\theta}^{\text{2stage}}_c = \hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,\alpha}(\hat{\theta}^{\text{global}}_c) \tag{12}$$

In sum, our GENet estimators are used for two-step estimation of local-and-global models. These transfer learning methods first make an initial estimate using data from abroad, and then use this estimate as a starting point for local estimation in any specific country.

In this connection, we denote the parameter ν as the "transfer parameter." Indeed, ν controls how strongly GENet shrinks toward the initial guess. If $\nu = 1$, we have full transfer since the GENet shrinks the parameter estimate toward the first-step estimate from abroad.

If $\nu \in (0, 1)$, we have partial transfer, where a higher ν means more information is transferred. Finally, $\nu = 0$ transfers no information, reducing to the standard ENet.

Tuning ν helps manage the bias–variance tradeoff. A higher ν uses more external information, potentially reducing bias but increasing variance if the guess is noisy. Since ENet is a special case, an optimally chosen ν allows GENet to outperform ENet. We provide formal conditions for this improvement in Section 3.4.

3.3 Non-Linear Transfer Learning

So far, we have focused on the linear model (4), which serves as a natural benchmark and allows us to establish strong theoretical results in Section 3.4. We now show how transfer learning can in fact implemented with *any* machine learning (ML) method, including non-linear ML.

Clearly, the single-region models in Section 3.1 can be generalized by replacing the ENet with another ML prediction method. Likewise, the GNet transfer learning methods from Section 3.2 can be generalized. To see how, it is useful to reinterpret our GENet by re-parametrizing (10) such that the GENet estimator is written as

$$\hat{\theta}^{\text{GENet}}_{c,\nu,\alpha,\lambda}(\hat{\theta}_0) = \nu\hat{\theta}_0 + \underset{\tilde{\theta}\in\mathbb{R}^p}{\arg\min}\{(Res_c - X_c\tilde{\theta})'(Res_c - X_c\tilde{\theta}) + P_{\lambda,\alpha}(\tilde{\theta})\} \tag{13}$$

where $\hat{\theta}_0$ is the first-stage estimate and $Res_c = Y_c - X_c\nu\hat{\theta}_0$ is the corresponding vector of residuals. This representation shows that the GENet estimator can be computed as follows:

1. Obtain an initial estimate $\hat{\theta}_0$ from another country (soft transfer) or from global data (two-stage).

2. Obtain local effect $\tilde{\theta}$ by computing residuals $Res_c = Y_c - X_c\nu\hat{\theta}_0$ for country c and regressing these residuals on X_c using a standard elastic net.

3. Form the final estimate $\hat{\theta} = \nu\hat{\theta}_0 + \tilde{\theta}$.

The same logic extends beyond linear regression and elastic net. Transfer learning can be implemented with *any* combination of ML methods as follows:

1. Obtain an initial prediction function, $\hat{f}$, based on another data set c' (from another country or global data) using any ML method, that is, $Y_{c'} = \hat{f}(X_{c'}) + \varepsilon_{c'}$.

2. Obtain a prediction function for local effects, $\hat{g}_c$, by computing residuals $Res_c = Y_c - \nu\hat{f}(X_c)$ for country c, and predicting these residuals using any ML method, that is, $Res_c = \hat{g}_c(X_c) + \epsilon_c$.

3. Form the final prediction as the sum of transferred knowledge and local correction, $\hat{Y}_c = \nu\hat{f}(X_c) + \hat{g}_c(X_c)$.

We apply this method empirically in Section 5.6 using XGBoost or random Fourier features.

3.4 Transfer Learning Benefits: GENet Prediction Precision

We wish to establish that our transfer learning methods reduce prediction errors. To analyze prediction errors, we consider the "mean squared error" (MSE) of any estimator $\hat{\theta}_c$ of the true parameter, $\theta_c = \theta_g + \theta_{\ell,c}$, in any country c for the linear models in (4). The MSE is defined as

$$MSE(\hat{\theta}_c) = \mathbb{E}(\|\hat{\theta}_c - \theta_c\|_2^2) \tag{14}$$

In other words, we consider the expected size of the difference between the estimated parameter (which is random because of estimation error) and the true parameter (which is also random in our Bayesian framework). As is well-known, this MSE captures the expected prediction error. For example, when predicting the out-of-sample return on any stock with characteristic vector x, we use the prediction $\hat{\theta}_c'x$. This prediction has an expected squared error given by

$$\mathbb{E}(r_{i,c,t+1} - \hat{\theta}_c'x)^2 = \mathbb{E}(\theta_c'x + \varepsilon_{i,c,t+1} - \hat{\theta}_c'x)^2 = \sigma_\varepsilon^2 + \mathbb{E}(x'(\theta_c - \hat{\theta}_c))^2, \tag{15}$$

since $\varepsilon_{i,c,t+1}$ is independent of the other variables when the return is out of sample. To focus on the quality of the estimated model parameters $\hat{\theta}_c$ without reference to the specific stock characteristics x, we can consider the prediction error for the worst-case x (leaving out the constant term σ_ε^2)

$$\mathbb{E}\left(\sup_{x:\|x\|\leq 1}(x'(\theta_c-\hat{\theta}_c))^2\right)=\mathbb{E}(\|\hat{\theta}_c-\theta_c\|_2^2)=MSE(\hat{\theta}_c), \tag{16}$$

which illustrates why having a low $MSE(\hat{\theta}_c)$ is a standard measure of expected prediction success.

In order to compare the MSE of the estimation methods from Section 3.2, we consider a Bayesian modeling framework with three natural assumptions. These assumptions clarify the distributions of the observations, $Y_{\mathcal{C}}=(Y_1',...,Y_C')'$, the parameters, $\Theta=(\theta_g,\theta_{\ell,1},\ldots,\theta_{\ell,C})$, and the characteristics, $X_{\mathcal{C}}=(X_1',...,X_C')'$, respectively:[4]

Assumption 1 *Conditional on Θ and $X_{\mathcal{C}}$, the observations $Y_{\mathcal{C}}$ are generated by (4), where $\varepsilon_1,\ldots,\varepsilon_C$ are independent and identically distributed with $\mathbb{E}(\varepsilon_c)=0$ and $\mathrm{Var}(\varepsilon_c)=\sigma_\varepsilon^2 I_{N_c}$.*

Assumption 2 *Conditioned on $X_{\mathcal{C}}$, the parameters $\theta_g,\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are independent zero-mean such that $\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are identically distributed, $\mathrm{Var}(\theta_g)=\sigma_g^2 I_p$, $\mathrm{Var}(\theta_{\ell,c})=\sigma_\ell^2 I_p$, where $\sigma_g^2>0$ and $\sigma_\ell^2\geq 0$.*

Assumption 3 *The variables $X_1,\ldots,X_C$ are identically distributed such that X_c has full column rank with probability 1, $\mathbb{E}\|(X_c'X_c)^2\|<\infty$, and $X_1,\ldots,X_C,\theta_g,\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are independent.*

To be able to state our propositions precisely, we clarify the notation specifying our estimators such that their dependence on the penalty parameters is explicit. Further, for simplicity, we only consider the ridge-based estimators, meaning that all estimators are defined with the elastic net parameter $\alpha=0$ in all steps of the estimation procedure. In this case, closed-form solutions exist for our transfer learning estimators, as seen in Lemma 2 in Appendix B.

To understand our straightforward notation, consider for example $\hat{\theta}^{\text{2stage}}_{c,\nu,\lambda,\lambda_0}$, which has a subscript for the country c and three additional parameters that should be interpreted as

[4] Recall that N_c is the number of securities in country c, I is the identity matrix, and (of relevance for Assumption 3) the norm for a matrix $A=(a_{i,j})\in\mathbb{R}^{p\times p}$ is defined as $\|A\|=\sum_{i,j}|a_{i,j}|$.

follows. In the initial step, all observations $Y_{\mathcal{C}}$ are used to estimate the global component using the ridge penalty λ_0. In the second step, the global estimate in inserted into the GENet (10) with penalty parameters ν and λ using data from the country of interest, Y_c.

We are ready to state our main theoretical results. The proofs are deferred to Section B in the appendix.

Proposition 1 (Transfer beats local) *For any $\lambda, \lambda_0 > 0$, it holds that*

(i) there exists $\nu' > 0$ such that $MSE(\hat{\theta}^{soft}_{c,\nu,\lambda,\lambda_0}) < MSE(\hat{\theta}^{local}_{c,\lambda})$ for all $\nu \in (0, \nu']$, under Assumptions 1–2;

(ii) for all $\nu \in (0, 2)$ we have $MSE(\hat{\theta}^{2stage}_{c,\nu,\lambda,\lambda_0}) < MSE(\hat{\theta}^{local}_{c,\lambda})$, when the number C of countries is large enough, under Assumptions 1–3;

(iii) $\nu = 1$ minimizes $MSE(\hat{\theta}^{2stage}_{c,\nu,\lambda,\lambda_0})$ among all ν as $C \to \infty$, under Assumptions 1–3.

To understand the first part of the proposition, note first that the soft estimator seeks to improve on a local estimator by pulling it toward ν times a first-stage estimate of the global parameter. Using global data adds information and may reduce bias (lowering MSE), but also adds variance from estimation noise from the first-stage (increasing MSE). Therefore, using the first-stage estimator as reference point (corresponding to $\nu = 1$) may not beat the local estimator if the first-stage is very noisy. Nevertheless, Proposition 1(i) shows that using at least *part* of the global information is always a good idea, corresponding to $\nu \in (0, 1]$. Using a small amount of global information makes a first-order improvement in estimation precision and a second-order increase in variance, thus reducing MSE, which is why we introduced the ν parameter in the first place.

Proposition 1(ii)–(iii) show that, when the number of countries is large, the 2stage estimator always beats the local estimator. Intuitively, this result is driven by the ability of the 2stage estimator to make a precise estimate of the global parameter in the first stage when there are many countries to learn from — so using this precise estimate is always helpful for all $\nu \in (0, 1]$, and even for less natural choices, $\nu \in (1, 2)$. In the limit as the number of countries grows, the first-stage estimate gets more and more precise, so the first-stage estimate becomes the optimal reference point, corresponding to $\nu = 1$.

The fact that the 2stage estimator, when the number of countries increases, gives more and more precise estimates of the global parameter in the first stage is formulated theoretically in the following proposition. Additionally, the proposition demonstrates that, in the limit, the 2stage estimate is the generalized ridge estimate $\hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,0}(\theta_g)$ based on observations from country c and on transfer learning using θ_g.

Our next result shows that the soft transfer methods (soft and 2stage) not only beat the local estimator but can also beat the corresponding hard transfer methods (hard and global).

Proposition 2 (Soft transfer beats hard) *When local effects are non-zero, $\sigma_\ell^2 > 0$, Assumptions 1-2 hold, and $\lambda_0 > 0$ small enough, it holds that*

(i) there exists λ' such that $MSE(\hat{\theta}^{soft}_{c,1,\lambda,\lambda_0}) < MSE(\hat{\theta}^{hard}_{c,\lambda_0})$ for all $\lambda > \lambda'$;

(ii) there exists λ' such that $MSE(\hat{\theta}^{2stage}_{c,1,\lambda,\lambda_0}) < MSE(\hat{\theta}^{global}_{c,\lambda_0})$ for all $\lambda > \lambda'$.

The proposition compares the hard estimators with the soft estimators in the case in which the two are similar in the sense that the soft estimators use the hard ones as reference point (corresponding to $\nu = 1$). In this case, using a least a small amount of local information (achieved via a large penalty λ on local parameters) reduces MSE relative to the hard estimators that use no country-specific information.

3.5 Joint Modelling

Recall that our overall goal is to estimate the local-and-global model of (4). Section 3.2 introduced estimators for a country's parameter $\theta_c = \theta_g + \theta_{\ell,c}$, in which a first-stage estimate using one or more other countries was combined with a second-stage estimate using data from the country of interest c. This section introduces an alternative approach, where we estimate parameters for all countries at the same time. This joint approach is related to the field of multitask learning within machine learning, see Caruana (1998). Its Bayesian interpretation also relates to hierarchical models, see Gelman and Hill (2006).

We estimate the collection of all parameters $\Theta = (\theta_g, \theta_{\ell,1}, \dots, \theta_{\ell,C})$ by the following joint minimization problem:

$$\hat{\Theta} = \arg\min_{\Theta} \Big\{ \sum_c (Y_c - X_c(\theta_g + \theta_{\ell,c}))'(Y_c - X_c(\theta_g + \theta_{\ell,c})) + P_{\lambda_g,0}(\theta_g) + \sum_c P_{\lambda_{\ell,c},0}(\theta_{\ell,c}) \Big\}. \quad (17)$$

Here $P_{\lambda_g,0}, P_{\lambda_{\ell,1},0}, \ldots, P_{\lambda_{\ell,C},0}$ are penalization functions of the type (6), each with their own penalty parameters. Due to the complexity of the joint minimization in (17), we only use ridge based estimators, that is $\alpha = 0$ in (6). With this type of penalization functions, the minimization problem becomes

$$\hat{\Theta} = \arg\min_{\Theta} \Big\{ \sum_c (Y_c - X_c(\theta_g + \theta_{\ell,c}))'(Y_c - X_c(\theta_g + \theta_{\ell,c})) + \lambda_g \theta_g' \theta_g + \sum_c \lambda_{\ell,c} \theta_{\ell,c}' \theta_{\ell,c} \Big\}. \quad (18)$$

The idea behind using a joint model instead of the country-specific approaches from Section 3.2 is that information from all countries, and the relations between them, are included in the estimation. The disadvantage of the joint approach is the substantial increase in complexity when estimating all parameters at the same time and the penalty parameters $\lambda_g, \lambda_{\ell,1}, \ldots, \lambda_{\ell,C}$ must be tuned jointly, which can lead to more numerical problems than the estimators in Section 3.2 and renders the use of elastic net penalty functions too computationally demanding in many practical applications.

To formulate our closed-form solution, we need to introduce some extra pieces of notation. We collect all local parameters in $\theta_\ell = (\theta_{\ell,1}', \ldots, \theta_{\ell,C}')'$ and define the diagonal matrix of penalties $\Lambda_\ell = \mathrm{diag}(\lambda_{\ell,1}, \ldots, \lambda_{\ell,C}) \otimes I_p$. Further, we introduce the auxiliary matrix $M = I_N - Z(Z'Z + \Lambda_\ell)^{-1}Z'$, where $N = N_1 + \cdots + N_C$ is the total number of observations across countries and Z is the block diagonal matrix with $X_1, \ldots, X_C$ on the diagonal. We are ready to state our joint estimator.

Proposition 3 *The unique solution $\hat{\Theta} = (\hat{\theta}_g, \hat{\theta}_\ell)$ to the joint optimization (18) is*

$$\hat{\theta}_g = (X_{\mathcal{C}}' M X_{\mathcal{C}} + \lambda_g I_p)^{-1} X_{\mathcal{C}}' M Y_{\mathcal{C}}, \quad (19)$$

$$\hat{\theta}_\ell = (Z'Z + \Lambda_\ell)^{-1} Z'(Y_{\mathcal{C}} - X_{\mathcal{C}} \hat{\theta}_g). \quad (20)$$

The corresponding total prediction parameter, $\hat{\theta}_c^{joint} = \hat{\theta}_g + \hat{\theta}_{\ell,c}$, for any country c is

$$\hat{\theta}_c^{joint} = (X_c' X_c + \lambda_{\ell,c} I_p)^{-1}(X_c' Y_c + \lambda_{\ell,c}\hat{\theta}_g). \tag{21}$$

The proposition provides a closed-form solution for our joint estimator. Further, the proposition shows that even the joint estimator can be viewed as a two-stage estimator. In the first step, the global parameter is estimated as (19). In the second step, then either all the local parameters are estimated as (20), or, more simply, each country's total prediction parameter is estimated as (21).

Conferring with the definitions of the generalized ridge estimator from (A.1) and the 2stage estimator, we see a strong resemblance between the joint estimator and the 2stage estimator: The 2stage estimator is an GENet estimator that uses observations from country c and transfers using $\hat{\theta}_c^{\text{global}}$, whereas the joint estimator is a generalized ridge estimator (a special case of the GENet) that transfers using $\hat{\theta}_g$.

While the joint estimator $\hat{\theta}_c^{\text{joint}}$ decomposes like a two-stage estimator, the penalty parameters $\lambda_{\ell,1}, \ldots, \lambda_{\ell,C}$ must be chosen jointly in the tuning stage. Indeed, each local penalty parameter $\lambda_{\ell,c}$ not only affects the country's joint estimator joint estimator $\hat{\theta}_c^{\text{joint}}$ directly through (21), it also affects the global parameter $\hat{\theta}_g$ in (19) since it is part of the M matrix.

Given the similarity to the 2stage estimator, we can similarly show that the joint estimator delivers lower expected prediction errors than a local model, under certain conditions. To obtain these results, we need some regularity on the penalty parameters:

Assumption 4 *The penalty parameters $\lambda_{\ell,1}, \ldots, \lambda_{\ell,C}$ are deterministic and equal.*

In the empirical exercise, we allow different penalty parameters in different countries. Such differences are allowed in the more general Assumption 5, stated in appendix. Either way, we have the following result on the prediction power of the joint estimator.

Proposition 4 (Joint beats local) *Under Assumptions 1–4 (or Assumption 5),*

(i) $\hat{\theta}_g \to \theta_g$ and $\hat{\theta}_c^{joint} \to \hat{\theta}_{c,1,\lambda,0}^{GENet}(\theta_g)$ as $C \to \infty$, both almost surely and in L^2.

(ii) in the case of non-zero local effects, $\sigma_\ell^2 > 0$, $MSE(\hat{\theta}_c^{joint}) < MSE(\hat{\theta}_c^{local})$, when the number C of countries is large enough.

Proposition 4 shows that when the number of countries increases, the joint estimator converges to the generalized ridge estimator based on transfer learning using θ_g. This is the same limit as for the 2stage estimator found in Lemma 3 in Appendix B. Therefore, similarly to the 2stage estimator, the joint estimator is superior to the local estimator in the limit of an increasing number of countries.

Finally, we note that the closed-form solution also has the following natural Bayesian interpretation. Suppose that we assume the priors

$$\theta_g \sim N\left(0, \sigma_g^2 I_p\right), \quad \theta_{\ell,c} \sim N\left(0, \sigma_{\ell,c}^2 I_p\right) \tag{22}$$

where $\sigma_g^2 = \sigma_\varepsilon^2/\lambda_g$ and $\sigma_{\ell,c}^2 = \sigma_\varepsilon^2/\lambda_{\ell,c}$ for $c = 1, \ldots, C$, and that the conditional distribution of $Y_{\mathcal{C}}$ given $(\theta_g, \theta_\ell, X_{\mathcal{C}})$ is as described in Assumption 1, assuming additionally that each ε_c is normally distributed. Then the estimators $(\hat{\theta}_g, \hat{\theta}_\ell)$ from Proposition 3 are the most likely parameters in the posterior distribution given the observed data:

$$(\hat{\theta}_g, \hat{\theta}_\ell) = \underset{\theta_g, \theta_\ell}{\arg\max}\, f(\theta_g, \theta_\ell \mid Y_{\mathcal{C}}, X_{\mathcal{C}}). \tag{23}$$

Details are found in Section C.2 of Appendix C. In this Bayesian interpretation, the joint model offers a generalisation of the mean–variance structure specified by Assumptions 1-3 since the variables $\theta_{\ell,c}$ are allowed to have different marginal variances in different countries.

4 Data and Methodology

4.1 Data

We use CRSP for US return data and Compustat for stock returns outside the US as well as for accounting data. All variables are in US dollars using Compustat exchange rates. For each country, we get the return of the local market portfolio, $R_{c,t}^m$, from Jensen et al. (2023) as well as the corresponding excess return, $r_{c,t}^m = R_{c,t}^m - R_t^f$, after subtracting the risk-free rate, R^f, measured as the US Treasury bill rate. We compute a global market return as the value-weighted average of the local market returns in the countries in our sample.

For each stock i in country c with total return $R_{i,c,t}$ at time t, we compute its return net of the cross-sectional average return, $r_{i,c,t} = R_{i,c,t} - \bar{R}_{c,t}$, where $\bar{R}_{c,t} = \frac{1}{N_{c,t}} \sum_i R_{i,c,t}$ is the average return in country c. We compute the return in this way (rather than computing returns in excess of the risk-free rate) in order to remove common shocks to each local market and focus on predicting stocks' relative outperformance or underperformance (similar to time-country fixed effects, or random effects in statistics).

We use the 115 firm characteristics that underlie the well-known equity factors studied in Jensen et al. (2023),[5] except for characteristics that require more than 5 years of data (e.g., years 6-10 lagged returns) to ensure enough coverage in all countries (as in Jensen et al. 2022), resulting in 111 characteristics. We rank the characteristics cross-sectionally within each country and then normalize them to be between -1 and 1. For much of our analysis, we split the data into training and test periods, where the former consists of all data up until the end of 1999 and the latter is 2000-2021. For a country to be included in our sample, we require more than 10 years of data in the training period, except that we also include China due to its large cross-section even though it only starts in 1993. Based on this method, our sample includes 34 countries. To alleviate the effect of errors in Compustat returns, we trim returns at a 0.1% and 99.9% level cross-sectionally as in Jensen et al. (2023).

4.2 Cross-Validation Methodology

Each method has a set of hyperparameters, which are chosen based on 10-fold cross-validation with MSE as the validation metric. For example, the GENet has the shrinkage parameters, λ, λ_c, the L^1 ratio, α_c, and the transfer parameter, ν_c.

We construct the folds such that the first fold consists of the first tenth of cross-sections, the second fold consists of the second tenth of cross-sections, etc., as in Kelly and Xiu (2023). For the joint model, the folds are constructed slightly differently: Fold 1 contains all data from the first month of the training period, the 11th month, and so on for every tenth month. Fold 2 then contains all data from the second month, 12th month, and so on for all 10 folds. This producedure for the joint model ensures that every fold contains roughly the same

[5] Characteristics are generated using the code from https://github.com/bkelly-lab/ReplicationCrisis/tree/master/GlobalFactors.

number of cross-sections from every country (i.e. the same number of US cross-sections in each fold, the same number of Danish cross-sections in each fold, etc.) despite the different sample starting points across countries.

We use Bayesian hyperparameter optimization to decide on a set of hyperparameters using the Python Optuna package (Akiba et al., 2019). This optimization method involves using techniques from online learning to learn from the validation scores of different sets of hyperparameters to determine the next combination of hyperparameters to test.[6]

The L^1 ratio and the transfer parameter, ν, are naturally restricted to be between 0 and 1, while λ_c only needs to be positive. We search for a λ_c in log-space, concretely $\lambda_c \in 10^{[-3,5]}$, where we optimize for the exponent. We let the Bayesian optimizer run for 100 iterations, searching for an optimal combination of hyperparameters. Appendix E shows the spaces over which we tune the hyperparameters for all models considered.

The joint model is harder to tune and requires more care. We take the tuned shrinkage for the purely global model as the penalty on the global component, λ_g. We then use the Bayesian optimization method to determine the optimal shrinkage λ_c for a 2stage model for country c with $\nu_c = 1$ and an L^1 ratio of 0, i.e. only using an L^2 penalty. This version of the 2stage model, being closest to the joint model, provides a reasonable tuning of the hyperparameters. Attempting to tune all 35 hyperparameters simultaneously runs into the curse-of-dimensionality and yields much worse results.

The reason we use Bayesian hyperparameter optimization is due to the optimal shrinkage varying by country by orders of magnitude. The combination of having many hyperparameters and some of them having optimal values in very different ranges makes grid search infeasible. This is especially important for the joint model due to having to tune 35 hyperparameters simultaneously. Additionally, this process makes it simple to tune hyperparameters across each country in parallel.

[6] We use the default parameters for the Optuna package. It uses the Tree-structured Parzen Estimator to fit two Gaussian Mixture Models, modeling the distribution of validation scores conditional on the hyperparameters. It uses these probabilistic models to determine the hyperparameters with the largest expected improvement over the current best validation score. Those hyperparameters are then used in the next trial.

4.3 Model-Comparison Methodology: R^2 and Portfolios

To compare the predictive performance of two models, we compare their out-of-sample R-squared, following Gu et al. (2020)

$$R^2_{OOS} = 1 - \frac{\sum_{t=1}^{T}\sum_{c=1}^{C}\sum_{i=1}^{N_{c,t}}(r_{i,c,t} - \hat{r}_{i,c,t})^2}{\sum_{t=1}^{T}\sum_{c=1}^{C}\sum_{i=1}^{N_{c,t}} r_{i,c,t}^2}. \tag{24}$$

We can also compare models in each country by their country-R^2_{OOS}

$$R^2_{OOS,c} = 1 - \frac{\sum_{t=1}^{T}\sum_{i=1}^{N_{c,t}}(r_{i,c,t} - \hat{r}_{i,c,t})^2}{\sum_{t=1}^{T}\sum_{i=1}^{N_{c,t}} r_{i,c,t}^2}. \tag{25}$$

We form trading strategies based on their predictions to compare models by the economic value of their forecasts. For each model, we first sort stocks in each country based on the model's predicted excess returns, $\hat{r}_{i,c,t}$. We then construct a long-short factor in each country by going long the top third stocks with the highest predicted returns while shorting the bottom tercile. Within the long and short portfolios, stocks are weighted using the capped-value-weighted method of Jensen et al. (2023), meaning that stocks are value-weighted using weights that are capped such that the largest stocks do not receive an enormous weight while tiny stocks do receive tiny weights. Finally, we create an overall factor, f_t, by aggregating across countries, weighting each country by the sum of its capped market equities.

We can then evaluate each model factor's performance by regressing its excess returns on different control variables:

$$f_t = \alpha + \beta^m r_t^m + \beta^{control} control_t + \varepsilon_t \tag{26}$$

The intercept is denoted the alpha, α, and the alpha divided by the residual volatility is denoted the information ratio, $IR = \alpha/\sigma(\varepsilon_t)$. For example, if the only right-hand-side variable is a constant, then alpha equals the average excess return, and the IR is the same as the Sharpe ratio. If the market return is also on the right-hand side, the alpha is the CAPM alpha.

5 Empirical Results

We first want to analyze whether a global element to asset pricing exists. Said differently, we test (and reject!) the null hypothesis that asset pricing has *no* global element.

5.1 Is There a Global Element to Asset Pricing?

If there is no global element, $\theta_g = 0$, then the natural estimation strategy is to fit a model for each country using only that country's own data, i.e., a local model. In contrast, a model trained in one country should not work in other countries if there truly is no global component.

We first consider whether parameters estimated on US data can be transferred to other countries. For this analysis, we estimate a model on all US data up until 1999 (hard transfer model) or on data from each country outside the US until 1999 (local models), and then evaluate the performance over the test period 2000-2021 for each country.

Figure 1 plots the R^2_{OOS} for each country for, respectively, the local models and the hard transfer model. The countries are sorted by the amount of training data available, starting with the countries with the most data.

In most countries, the local models have positive R^2_{OOS}, which is evidence that the local models have some predictive ability. In countries with little data, such as Ireland or Argentina, the local model is not able to learn anything with the limited amounts of data (R^2_{OOS} close to zero). On the other hand, the hard transfer model performs well in those countries and outperforms the local models in most countries, even those with more data.

The left-most bar is the average R^2_{OOS} across all countries for local and hard transfer models. The overall R^2_{OOS} for the collection of local models is 0.104% while that of the hard transfer model is 0.205%.

If there is no global component, the hard transfer model shouldn't beat a naive benchmark of predicting an excess return of 0. If the local models have positive R^2_{OOS}, their estimates of each θ_c will be better than those of the naive estimator. Obviously, the local models then pose an even tougher challenge for the hard transfer models.

In other words, the fact that hard transfer outperforms the local models when it comes to

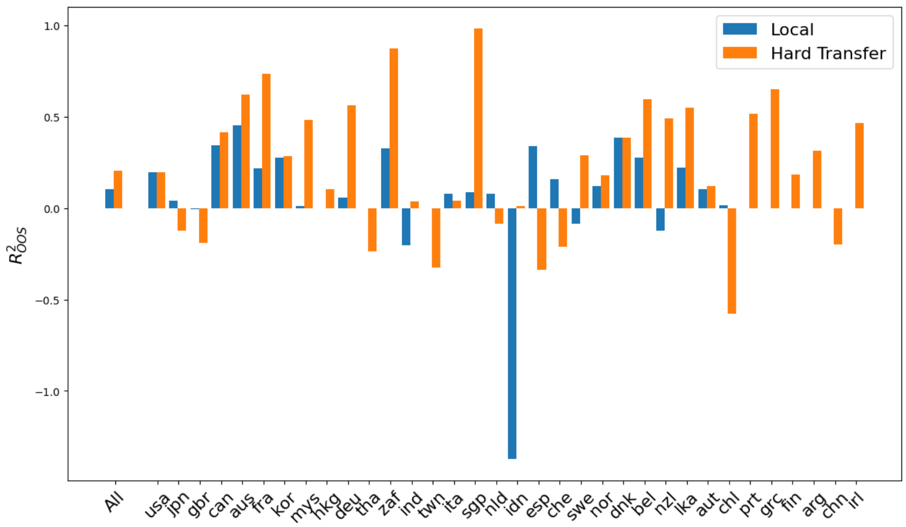

Figure 1: **Performance of Local and Hard Transfer Models.** This figure plots the average R^2_{OOS} (measured in percent) for each country based on the out-of-sample test period 2000-2021 for, respectively, a local and a hard transfer model. Countries are sorted by the amount of available training data. The local model is trained on the country's own data up until 1999. The hard transfer model is trained on US data up until 1999. 'All' refers to the average across countries, and the corresponding bars show that the local model has a positive R-squared and the hard transfer model works even better.

prediction, is evidence of a significant global component. Next, we test whether this superior prediction performance is exploitable in a trading strategy.

Figure 2 shows the Sharpe ratio of the long-short factors in each country constructed based on the models' predictions. Countries are sorted by the Sharpe ratio of their factor created using the local model. Even though the hard transfer model has never "seen" data from any of those countries, the factors constructed using that model outperform factors constructed using the local models in almost every country. Remarkably, to trade stocks in a country outside the US using the models that we consider, you would have been better off training a model on US data than training a model on that country's own data.

Aggregating these factors across countries, Table 2 shows the performance of the local and hard transfer for all stocks outside the US. We see that the local factor achieves significant average returns (first column) and CAPM alpha (second column) on its own. However, as soon as we add the hard factor to the regression, the alpha completely disappears (third column).

Turning to the performance of the hard transfer factor, we see a much higher average

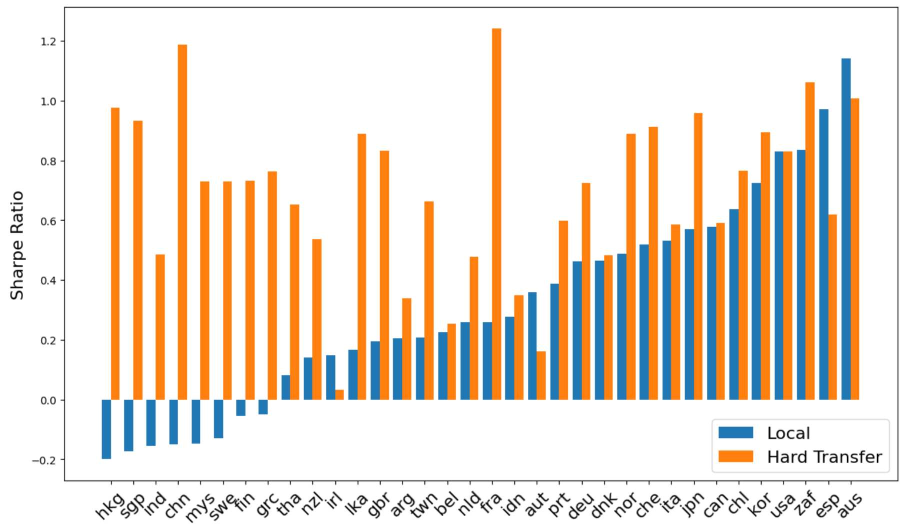

Figure 2: **Sharpe Ratio of Factors using Local vs. Hard Transfer Models.** This figure plots, for each country outside the US, the Sharpe ratio of a factor constructed using a model trained using the country's own data (local) and a model trained on US data (hard transfer), respectively. The countries are sorted by the Sharpe ratio of the factor constructed using the local model. The training period is up until 1999, and the test period is 2000-2021.

return (fourth column), higher alpha to the market (fifth column), and even a highly significant alpha when controlling for the local factor (sixth column). These findings suggest that the local models have not found any meaningful country-specific information to take advantage of whereas the hard factor has found information from US data that is applicable globally.

To show that it is not the US that is especially informative, we do the same exercise the other way around: We train a hard transfer model on data pooled from all countries outside the US and compare its performance to that of a model trained only on US data.

Unfortunately, even when pooling the data from every non-US country in the training period until 1999, this global sample is still smaller than the US sample until 1999—since the US data starts in 1926 while the global data starts in the 1980s. This is a problem if we wish to show that with enough training data, a model trained on non-US data has comparable out-of-sample performance to a model trained using US data. To address this issue, we train the hard transfer model on data from non-US countries over 1982-2021 and train the local model on US data over the same time period 1982-2021. We then use the old US data 1926-1981 as the test sample.

	Local (Ex-US)			Hard (US)		
α	3.17***	3.14***	0.41	9.47***	9.46***	8.40***
	(2.94)	(2.90)	(0.36)	(8.09)	(8.07)	(7.40)
Mkt		0.01	0.01		0.00	-0.00
		(0.85)	(0.83)		(0.19)	(-0.07)
Hard (US)			0.29***			
			(5.32)			
Local (Ex-US)						0.34***
						(5.32)
IR	0.63	0.62	0.09	1.73	1.72	1.61

Table 2: **Local and Hard Transfer for World-Ex-US Stocks.** This table reports the alphas of factors trading stocks outside the US, constructed using models trained on, respectively, each country's local data (local) and US data (hard transfer). Each factor is constructed by ranking stocks by their predicted returns to form a long-short portfolio as in Jensen et al. (2023). The market portfolio is the global value-weighted portfolio. The training period is up until 1999, while the test period is 2000-2021. Alphas are annualized, and t-statistics are in parentheses. IR is the information ratio, computed as alpha divided by residual volatility.

We first evaluate these models using their R-squared. The hard (World-Ex-US) model achieves a strong predictive power with $R^2_{OOS} = 0.389\%$, while the local US model performs similarly with $R^2_{OOS} = 0.398\%$. Once again, this suggests the existence of a strong global component.

To analyze these findings in terms of portfolio performance, we create factors using the local (US) model as well as the hard transfer model (World-Ex-US) in the same manner as earlier. Table 3 shows the performance of each factor.[7] Both factors have large and significant average returns and CAPM alphas. The hard (World-Ex-US) factor even has a higher Sharpe ratio than the local (US) factor. Clearly, hard transfer performs well, again showing that the phenomenon was not unique to a model trained on US data. Instead, hard transfer works due to the larger amounts of training data as well as the importance of the global element of predictability.

Each model has alpha even when controlling for the other model, showing that US in-

[7]Given that the test period is before the training period, the factor returns could obviously not have been calculated ex-ante and, therefore, should not be viewed as a realistic trading simulation. The point of this exercise is simply to analyze how a model trained on data from non-US countries works on out-of-sample US data in comparison to a model trained on US data.

	Local (US)			Hard (World-Ex-US)		
α	12.57***	15.03***	6.40***	12.36***	12.47***	2.12**
	(9.65)	(13.15)	(7.26)	(10.93)	(10.93)	(2.35)
Mkt		-0.29***	-0.28***		-0.01	0.19***
		(-16.86)	(-22.58)		(-0.74)	(13.10)
Hard (Ex-US)			0.69***			
			(28.40)			
Local (US)						0.69***
						(28.40)
IR	1.12	1.54	0.91	1.27	1.28	0.3

Table 3: **Local and Hard Transfer Models for US Stocks.** This table reports the alphas of factors trading US stocks, constructed using models trained on, respectively, US data (local) and data for countries outside the US (hard transfer). Each factor is constructed by ranking stocks by their predicted returns to form a long-short portfolio as in Jensen et al. (2023). The market portfolio is the US value-weighted portfolio. The training period is 1982-2021 while the test period is 1926-1981. Alphas are annualized, and t-statistics are in parentheses. IR is the information ratio, computed as alpha divided by residual volatility.

vestors and asset pricers can also benefit from learning from global data. These findings provide further evidence of a large global component, but also hint at a local component that the local (US) model is capturing.

5.2 Is There a Local Element to Asset Pricing?

If asset pricing has no local element, then there is no reason to train individual models for each country. Instead, we should pool all available training to fit a single global model. This global model therefore serves as the benchmark to beat for models attempting to fit additional local components. In other words, if asset pricing has a local element, then methods using global and local data should outperform the purely global model.

Simply training a model on each country's own data (local) yields fairly poor results as we saw in Figure 1. However, such poor performance can arise due to the noise and small-sample problems in local estimates—even if a local component does exist. To be able to detect a local element using a more powerful method, we employ our local-and-global transfer learning methods. Intuitively, these estimators use knowledge of the global part of the signal to heighten its signal-to-noise ratio for detecting local components.

The local-and-global methods are soft, 2stage, and joint. All these three transfer models outperform the global model in terms of their predictive power. In particular, the global model has an overall R-squared of $R^2_{OOS} = 0.257\%$, which is increased to $R^2_{OOS} = 0.265\%$ for soft, $R^2_{OOS} = 0.289\%$ for 2stage, and $R^2_{OOS} = 0.263\%$ for joint. In other words, at least in this predictive sense, our transfer learning methods capture local components.

Next, we test whether this superior predictive performance translates into economic benefits in the form of superior performance of the corresponding trading strategies. We construct factors using the cross-sectional return predictions of, respectively, the soft, 2stage, and joint models as explained in Section 4.3. Table 4 reports their performance for the sample of all global stocks (Panel A) and all stocks outside the US (Panel B). In both panels, all these three factors have significant average returns and CAPM alphas, and all have high Sharpe ratios.

In both Panel A and Panel B, the soft and 2stage factors have higher and significant alphas to the global market, the global factor, and the local factor. The outperformance of the transfer learning models is stronger in Panel B (where US stocks are excluded) because this panel focuses on the stocks where the models have more different predictions. Said differently, the results in Panel A are dominated by US stocks, where the models are similar, which partly drowns out potential differences between the factors.

The joint factor performs worse, achieving no alpha when accounting for exposure to the market and the global factor. The joint model captures global and local elements simultaneously, which could in principle be helpful, but turns out to lead to worse performance than our other local-and-global models. There could be several reasons for this difference. First, the soft and 2stage models are more flexible due to the extra transfer parameter. Furthermore, due to a curse of dimensionality in hyperparameters, tuning for the joint model (tuning $C + 1$ hyperparameters simultaneously) is much more difficult than tuning the soft or 2stage models. For tractability, the joint model also only uses a Ridge penalty, while soft and 2stage use the more flexible elastic net.

To recap, Table 4 provides evidence of economic benefits of capturing both local and global elements using two of our transfer learning methods. Indeed, the soft and 2stage

Panel A: All Stocks

	Soft				2Stage				Joint			
α	9.76***	9.72***	1.00***	1.17***	9.83***	9.75***	0.91***	1.12***	9.70***	9.59***	0.33	0.36*
	(7.17)	(7.12)	(3.51)	(4.29)	(7.16)	(7.10)	(4.07)	(5.89)	(6.75)	(6.68)	(1.62)	(1.74)
Mkt		0.01	-0.02***	-0.02***		0.02	-0.01**	-0.01***		0.03	-0.00	-0.00
		(0.64)	(-5.10)	(-5.44)		(1.28)	(-2.42)	(-2.94)		(1.53)	(-0.98)	(-0.99)
Global			0.93***	0.84***			0.95***	0.83***			0.99***	0.98***
			(81.80)	(44.41)			(105.65)	(62.46)			(121.78)	(68.11)
Local				0.13***				0.16***				0.02
				(5.63)				(10.28)				(1.26)
IR	1.53	1.52	0.81	1.0	1.53	1.52	0.94	1.37	1.44	1.43	0.37	0.41

Panel B: Ex-US Stocks

	Soft				2Stage				Joint			
α	9.94***	9.95***	1.83***	1.83***	10.02***	10.00***	1.63***	1.64***	9.54***	9.47***	0.22	0.23
	(9.06)	(9.06)	(3.96)	(4.13)	(9.16)	(9.12)	(4.58)	(5.47)	(8.15)	(8.10)	(1.32)	(1.38)
Mkt		-0.01	-0.02***	-0.02***		0.01	-0.01*	-0.01**		0.02	0.00*	0.00*
		(-0.44)	(-4.25)	(-4.53)		(0.65)	(-1.88)	(-2.50)		(1.48)	(1.80)	(1.76)
Global			0.86***	0.83***			0.89***	0.83***			0.98***	0.97***
			(39.85)	(36.73)			(53.16)	(54.86)			(123.62)	(117.17)
Local				0.11***				0.17***				0.04***
				(4.65)				(10.45)				(4.40)
IR	1.93	1.94	0.95	0.99	1.95	1.95	1.09	1.31	1.74	1.73	0.32	0.33

Table 4: **Performance of Transfer Learning Models.** This table reports the performance of long-short portfolios formed based on global, soft transfer, and 2stage models. All numbers are annualized and in percent. The global model is a model trained by pooling all available training data (not accounting for possible country-specific differences). The soft model is trained by combining US data with local country-specific data. The 2stage model is trained by combining pooled global data with local country-specific data. The joint model estimates global and local components simultaneously. The training period is up until 1999 while the test period is 2000-2021. Panel A includes all stocks while Panel B includes only stocks outside the US.

methods outperform the purely global model as well as the combination of the global and local models.

To show the performance of our models in more detail, Table 5 reports the performance of decile portfolios of stocks sorted on each model's return predictions. Portfolio 1 consists of the stocks with the lowest predicted returns while portfolio 10 contains the stocks with the highest predicted returns. In each portfolio, stocks are capped-value-weighted within each country and across countries. The table shows that the average returns, CAPM alphas, and Sharpe ratios all increase with portfolio number, indicating that all the models have found meaningful conditional information in firm characteristics.

Table 5 also shows the performance of two long-short portfolios. First, we consider the 10-minus-1 portfolio that goes long the highest-expected-return portfolio while shorting the

Panel A: Average Return of Decile Portfolios

	1	2	3	4	5	6	7	8	9	10	10-1	*t*-stat	Tercile	*t*-stat
Local	0.69	4.02	6.18	6.90	7.81	8.31	8.38	8.89	10.38	12.54	11.85	5.79	3.17	2.94
Hard	-2.12	3.00	5.71	7.45	8.49	9.11	10.57	11.30	13.44	15.98	18.10	7.84	9.47	8.09
Global	-1.85	3.34	6.18	7.49	8.94	9.04	10.81	11.34	13.89	16.56	18.41	7.99	9.46	8.02
Soft	-2.12	3.29	5.34	7.33	8.77	9.45	10.65	11.60	13.53	16.82	18.94	8.50	9.94	9.06
2Stage	-1.84	3.34	6.25	7.27	8.98	9.23	10.96	11.86	13.96	16.95	18.79	8.32	10.02	9.16
Joint	-2.38	2.73	5.75	7.38	8.42	9.33	10.21	11.70	12.98	16.03	18.41	8.13	9.54	8.15

Panel B: CAPM Alpha of Decile Portfolios

	1	2	3	4	5	6	7	8	9	10	10-1	*t*-stat	Tercile	*t*-stat
Local	-0.79	2.65	4.83	5.54	6.48	6.96	7.02	7.49	8.94	10.99	11.79	5.76	3.14	2.90
Hard	-3.63	1.63	4.38	6.11	7.17	7.77	9.18	9.91	12.01	14.45	18.07	7.81	9.46	8.07
Global	-3.34	2.00	4.84	6.15	7.61	7.69	9.42	9.92	12.45	15.00	18.33	7.94	9.40	7.98
Soft	-3.63	1.90	3.99	6.00	7.46	8.11	9.27	10.21	12.11	15.29	18.92	8.47	9.95	9.06
2Stage	-3.33	1.98	4.91	5.93	7.66	7.88	9.58	10.47	12.52	15.39	18.72	8.28	10.00	9.12
Joint	-3.86	1.38	4.39	6.06	7.10	7.98	8.82	10.29	11.54	14.50	18.36	8.10	9.47	8.10

Panel C: Sharpe Ratio of Decile Portfolios

	1	2	3	4	5	6	7	8	9	10	10-1	Tercile
Local	0.04	0.24	0.37	0.41	0.47	0.50	0.49	0.50	0.56	0.62	1.24	0.63
Hard	-0.12	0.18	0.35	0.45	0.52	0.54	0.61	0.64	0.72	0.79	1.67	1.73
Global	-0.10	0.20	0.38	0.46	0.54	0.54	0.62	0.63	0.73	0.80	1.70	1.71
Soft	-0.11	0.19	0.33	0.45	0.54	0.56	0.62	0.66	0.73	0.84	1.81	1.93
2Stage	-0.10	0.20	0.38	0.44	0.54	0.55	0.63	0.67	0.74	0.83	1.77	1.95
Joint	-0.13	0.16	0.35	0.45	0.52	0.56	0.59	0.66	0.69	0.79	1.73	1.74

Table 5: **Performance of Decile Portfolios.** This table shows the performance of decile portfolios for stocks sorted by their predicted returns based on, respectively, local, hard, global, soft, 2stage, and joint models. In each portfolio, stocks are capped-value-weighted within each country and across countries. The '10-1' portfolio goes long the stocks with high predicted returns in portfolio 10 while shorting those with low predicted returns in portfolio 1. The tercile portfolios goes long stocks in portfolios 10, 9, and 8, and shorts stock in portfolios 1, 2, and 3. The training period is up until 1999 while the test sample is global stocks 2000-2021. US stocks are excluded. All numbers are annualized.

lowest one. Second, we also look at the tercile portfolios used in our factor construction above. As before, we see that the local model performs the worst, the global model performs well, and the soft and 2stage models outperform the global model in terms of average return, CAPM alpha, and Sharpe ratio. However, the extra performance achieved by soft and 2stage is lower than the difference in performance between purely local and global.

In summary, we find evidence of statistically significant, but economically more modest, local elements of asset pricing, captured using our transfer learning methods.

5.3 What is the Global Component?

To see which characteristics are the most important globally, we consider the predictive coefficients in the global model fitted using all stocks in the full sample period, 1926-2021. Further, we are also interested in whether these coefficients are stable across the world and over time. Therefore, we also consider the coefficients when the global model has been fitting on five non-overlapping sub-samples: US stocks in the periods 1926-1981, 1982-1999, and 2000-2021, as well as all non-US stocks in the periods 1982-1999 and 2000-2021.

Figure 3 plots the coefficients of the 20 most important features, measured by the absolute size of the coefficients estimated in the full sample. The coefficients are signed consistently with the paper they were originally introduced in, similar to Jensen et al. (2023). A positive coefficient thus means that the coefficient has the same sign as in the prior literature. However, we note that our predictive coefficients have a slightly different interpretation. Our coefficients are multivariate predictive coefficients, that is, each coefficient represents the predictive power of a signal *when holding all other characteristics constant*. In contrast, the literature looks at each signal by itself (or controlling for a few factors such as the Fama-French factors).

In the 20 most important features, we see some well-known ones like "ret_1_0" (short-term reversal, Jegadeesh 1990), "market_equity" and "be_me" (size and book-to-market, Fama and French 1992), as well as "ret_9_1" (9-month price momentum, Jegadeesh and Titman 2001). The 20 most important features belong to a range of different clusters, as defined by Jensen et al. (2023). The clusters with the most factors among the top 20 are "Low Risk", "Momentum", and "Size," followed by "Value" and several others.

A striking feature of Figure 3 is how similar the coefficients are across samples. Considering constant technological and cultural advances, the idea that characteristics predict stock returns in the same way across countries and time periods can at first seem far-fetched (Han et al., 2024). However, if societal changes merely affect the distribution of characteristics, then our multivariate regression coefficients may be stable even if univariate ones would not be.

Table 6 quantifies how similar the predictive coefficients are across the five non-overlapping

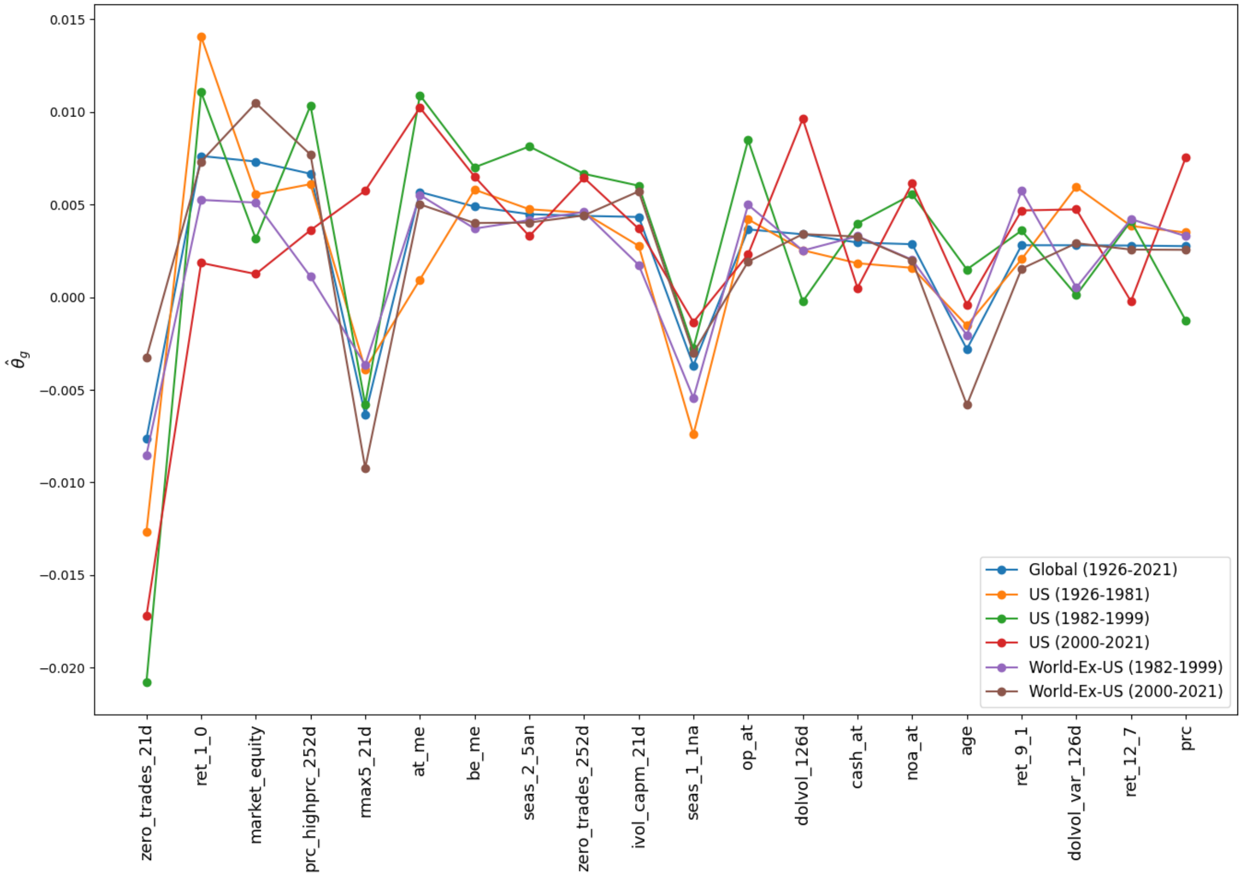

Figure 3: **20 Most Important Global Features.** This figure plots the 20 most important features of the global return-predicting component, measured by the absolute size of their coefficients, $|\theta_g|$. The figure reports the value of these coefficients estimated using six different samples: US data in 1926-1981, 1982-2000, and 2000-2021, all stocks outside the US in 1982-2000 and 2000-2021, as well as all stocks in the period 1926-2021. The features are ordered by their importance in all stocks in the period 1926-2021. The features are signed in the same way as the corresponding original papers (as in Jensen et al. (2023)).

samples, as well as with the total sample. Panel A shows the pairwise correlations between the six sets of estimated predictive parameters, $\hat{\theta}_g$. We see that the estimated coefficients are highly correlated, further underlining how consistent the estimates are across independent sub-samples.

Panel B and C of Table 6 show the robustness of this finding, illustrating the similarity across samples using alternative measures of similarity. In particular, Panel B addresses that the correlations in Panel A are affected by how features are signed, which is arguably arbitrary. To compute correlations in a way that does not depend on signing conventions,

Panel A: Correlation of Predictive Coefficients

	Global (1926-2021)	US (1926-1981)	US (1982-1999)	US (2000-2021)	World-Ex-US (1982-1999)	World-Ex-US (2000-2021)
Global (1926-2021)	1					
US (1926-1981)	0.80	1				
US (1982-1999)	0.79	0.70	1			
US (2000-2021)	0.62	0.52	0.63	1		
World-Ex-US (1982-1999)	0.74	0.64	0.61	0.39	1	
World-Ex-US (2000-2021)	0.91	0.62	0.58	0.42	0.62	1

Panel B: Cosine Similarity of Predictive Coefficients

	Global (1926-2021)	US (1926-1981)	US (1982-1999)	US (2000-2021)	World-Ex-US (1982-1999)	World-Ex-US (2000-2021)
Global (1926-2021)	1					
US (1926-1981)	0.81	1				
US (1982-1999)	0.81	0.71	1			
US (2000-2021)	0.65	0.54	0.65	1		
World-Ex-US (1982-1999)	0.75	0.65	0.63	0.42	1	
World-Ex-US (2000-2021)	0.91	0.63	0.60	0.45	0.63	1

Panel C: Correlation of Predictions

	Global (1926-2021)	US (1926-1981)	US (1982-1999)	US (2000-2021)	World-Ex-US (1982-1999)	World-Ex-US (2000-2021)
Global (1926-2021)	1					
US (1926-1981)	0.85	1				
US (1982-1999)	0.85	0.77	1			
US (2000-2021)	0.82	0.62	0.78	1		
World-Ex-US (1982-1999)	0.79	0.74	0.71	0.54	1	
World-Ex-US (2000-2021)	0.96	0.76	0.72	0.73	0.70	1

Table 6: **Similarity between Predictive Coefficients Estimated from Non-Overlapping Samples.** This table shows the similarity between predictive coefficients $\hat{\theta}_g$, estimated using a global model for each of five different samples: US data in 1926-1981, 1982-2000, and 2000-2021, all stocks outside the US in 1982-2000 and 2000-2021. The similarity is measured by pairwise correlation between predictive coefficients (Panel A), cosine similarity between predictive coefficients (Panel B), and correlation between predicted returns of all stocks in the period 1926-2021 (Panel C).

Panel B shows the cosine similarity. Cosine similarity is similar to correlation, but uses the prior mean 0 as opposed to the average of each $\hat{\theta}_g$.[8] Again, we find that the predictability parameters estimated using different sub-samples are very similar.

Since we are mostly interested in predictions, we also compute the pairwise correlation between predictions implied by each estimate, that is, $\hat{\theta}_g' x_{i,c,t}$, as reported in Panel C. Naturally,

[8]The signing of features affect the average coefficient, which in turn affects the correlations. If signing is arbitrary, then one could argue that the average in the correlation calculation should be zero, which is what is called cosine similarity, S:

$$S(\theta_1, \theta_2) = \frac{\theta_1' \theta_2}{\|\theta_1\|_2 \cdot \|\theta_2\|_2}. \tag{27}$$

since the different samples imply similar $\hat{\theta}_g$ estimates, these similar predictive coefficients in turn imply similar expected stock returns.

From Table 6 we especially note the similarity between parameters fitted using the total sample and each of the sub-samples. As one would expect, the estimates from the US samples are quite similar to the estimates from the total sample. The US is a substantial part of the global component. However, the estimates from the ex-US samples are also similar to those of the total sample. This finding suggests that the US is not playing a special role—large countries or regions are close to the global model because they have a lot of data to identify the underlying asset pricing model, which appears relatively global.

5.4 What Are the Local Components?

We are also interested in the coefficients that vary the most between countries. To consider these local coefficients, we fit the joint model on the sample 2000-2021 using the hyperparameters found in the training data.

Figure 4 shows the 20 features with the highest cross-country variance. In particular, the figure shows the variation of the total prediction parameter, $\hat{\theta}_c$, across countries. For each feature, the figure illustrates both the mean coefficient and two-standard-error bounds that reflect the cross-country variation. The features are sorted by the width of the bar, that is, the cross-country variation.

Interestingly, some of the most important characteristics in the global element are also the characteristics that vary the most across countries, such as size, short-term reversal, and assets-to-market. Many of the features with the highest cross-country variation fit into the "Value", "Low Risk", "Investment", or "Quality" clusters.

As an example, we consider the case of Japan in more detail. Japan is of special interest due to the discussion in the prior literature. For example, Fama and French (2012) find that momentum does not work in Japan, while Asness (2011) and Asness et al. (2013) argue the Japanese case could be partly due to noise and that the combination of value and momentum also has a higher Sharpe ratio than value alone in Japan. The latter finding implies that momentum may work when controlling for value, and our multivariate approach is ideal for handling such confounding effects. Figure 5 shows the predictive coefficients for the 20 most

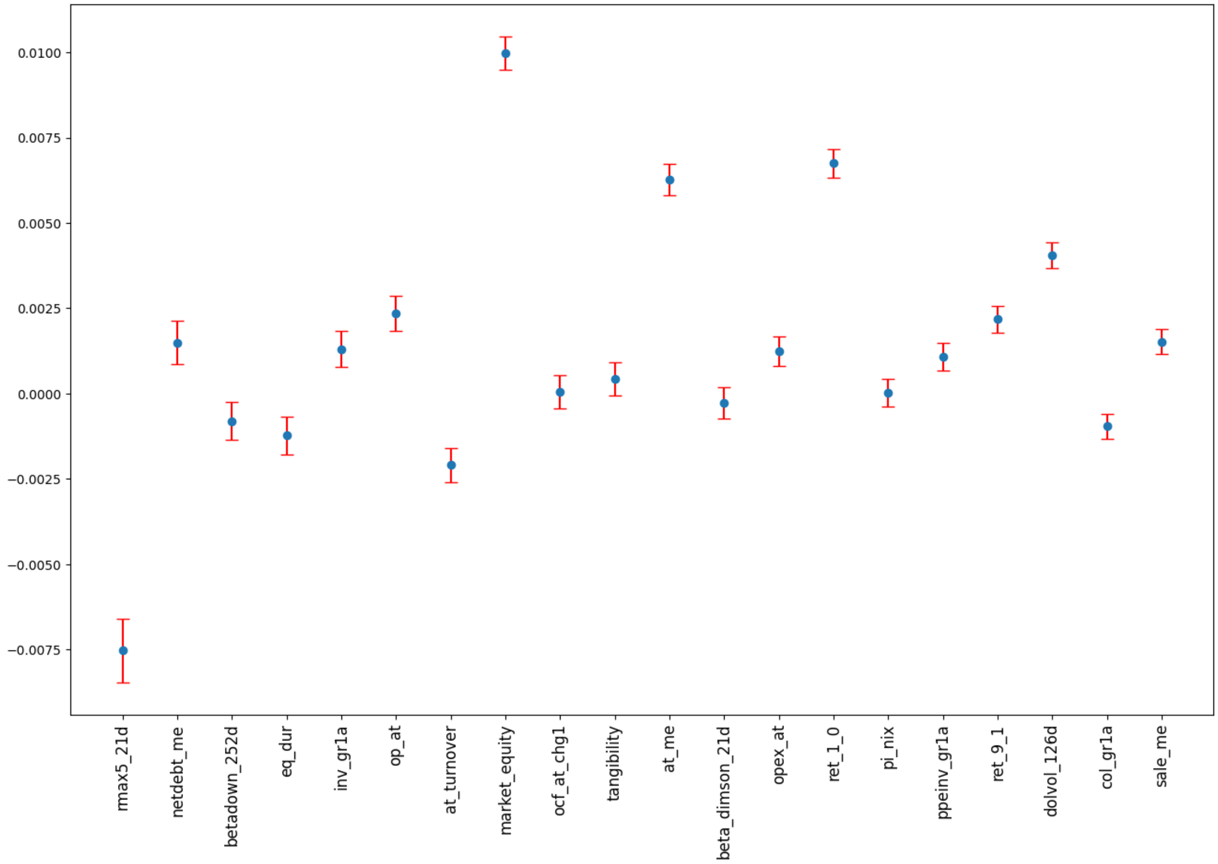

Figure 4: **The 20 Features with Highest Cross-Country Variation.** This figure plots the average return-predicting coefficient, $\theta_g + \theta_{\ell,c}$ across countries c, and a bar illustrating plus and minus two standard deviations across countries, showing the 20 coefficients with the highest cross-country variation. The joint model is estimated on a sample of global stocks 2000-2021.

important features in Japan in 2000-2021 and for comparison also shows the coefficients for other regions and samples. The figure shows that these multivariate coefficients estimated for Japan are very similar to estimates from the other samples.

So, while a country like Japan looks similar in terms of the multivariate predictive coefficients, other country differences could exist. For example, if Japan has a special constellation of characteristics, then univariate effects could look different even if multivariate ones look similar.

We note that we illustrate the local components using the joint model because the local components are immediately estimated using this method. The other local-and-global models—soft and 2stage—could also be considered, but the local component is not as clearly

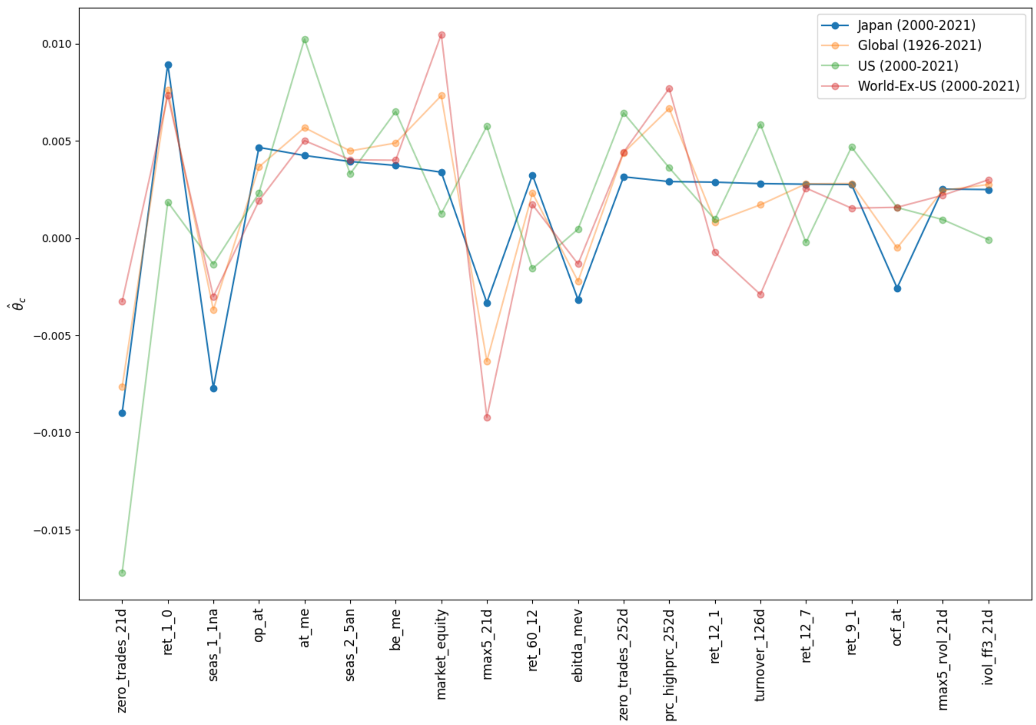

Figure 5: **20 Most Important Features in Japan.** This figure plots the 20 most important features for predicting stock returns in Japan, measured by the absolute size of their coefficients, $\left|\hat{\theta}^{\text{2stage}}_{\text{Japan}}\right|$. The figure reports the value of these coefficients estimated on Japanese stocks in 2000-2021 using our 2stage model, as well as US stocks in 2000-2021 using a purely local model, all stocks in 1926-2021, and all stocks outside the US in 2000-2021 using purely global models. Features are ordered by their importance in Japan. Coefficients are signed consistently with the paper, they were originally introduced in, similar to Jensen et al. (2023).

defined in general. The problem arises due to the transfer parameter, ν, used in these models. As an example, tuning hyperparameters for 2stage in Germany results in $\nu = 0.43$ and the penalized regression estimates of all the local coefficients are 0. In other words, $\hat{\theta}^{\text{2stage}}_{\text{Germany}} = 0.43\hat{\theta}^{\text{global}}$. The coefficients for Germany are then different from the purely global model, but the resulting sorted portfolios will be identical. More broadly, defining what the global and local components are is not straightforward whenever ν is different from 1, which is why we focus here on the joint model (which implicitly enforces $\nu = 1$).

In Appendix D we additionally study how the tuned transfer parameter ν varies across countries using soft transfer and 2stage. We see no clear effect of the number of training observations, except that ν in the 2stage model is slightly larger for countries with many observations in the training data.

5.5 Transferability of Mispricing vs. Risk

Our transfer learning methods are capable of picking up small local differences in the prediction function. A natural question is therefore: where do these local differences come from? For example, do they come from differences in risk pricing or behavioral mispricing? To address this question, we first separate characteristics into risk and mispricing, and then analyze which group has more local variation.

We categorize characteristics using the method of Chen et al. (2022), where a characteristic is classified according to whether its original authors propose it as risk, mispricing, or agnostic.[9]

Next, we estimate versions of our 2stage method, restricting local variation to a single type (risk, mispricing, or agnostic). Specifically, we start with the estimated global model, and then estimate the local-and-global model, allowing only one group of parameters to deviate from their values in the purely global model.[10]

Table 7 shows the results. In particular, it shows the performance of 2stage models, where the second stage is fitted using only mispricing, risk, or agnostic characteristics. Interestingly, column six shows that the model with local variation in risk-based characteristics has no alpha relative to the global model. This finding suggests that there is no local variation in risk pricing.

In contrast, we find significant alpha for the model with local variation in mispricing or agnostic factors. Hence, mispricing appears to have local elements, which could be driven by cultural differences in behavioral trading patterns, institutional features, or market structures.

One concern could be that we identify a local component of mispricing, not risk, because there are more characteristics classified as mispricing, yielding more statistical power. Indeed, among all characteristics, 56 are mispricing, 22 risk-based, and 33 agnostic. To address this potential issue, we partition the mispricing characteristics into three approximately equal-sized groups based on the clusters in Jensen et al. (2023), denoted as "Investment",

[9] Appendix F presents further details on the methodology and the complete classification of all characteristics.

[10] Equivalently, using the same re-parametrization as for non-linear transfer learning (13), we can choose to fit the local component $\tilde{\theta}$ with variation restricted to one group of characteristics.

Panel A: All Stocks

	Mispricing			Risk			Agnostic		
α	9.67***	9.59***	0.66**	9.39***	9.26***	-0.04	9.47***	9.37***	0.58**
	(6.92)	(6.86)	(2.28)	(6.54)	(6.47)	(-0.61)	(6.90)	(6.83)	(2.24)
Mkt		0.02	-0.01**		0.03*	0.00***		0.02	-0.00
		(1.21)	(-2.27)		(1.83)	(3.45)		(1.40)	(-1.36)
Global			0.96***			1.00***			0.94***
			(83.39)			(351.37)			(90.52)
IR	1.48	1.47	0.53	1.39	1.38	-0.14	1.47	1.46	0.52

Panel B: Ex-US Stocks

	Mispricing			Risk			Agnostic		
α	9.75***	9.72***	1.18**	9.34***	9.27***	-0.05	9.42***	9.39***	1.13***
	(8.51)	(8.47)	(2.55)	(7.95)	(7.90)	(-0.46)	(8.56)	(8.52)	(2.67)
Mkt		0.01	-0.01*		0.02	0.00***		0.01	-0.00
		(0.54)	(-1.73)		(1.51)	(2.91)		(0.81)	(-1.08)
Global			0.91***			0.99***			0.88***
			(41.66)			(186.69)			(44.38)
IR	1.81	1.81	0.61	1.69	1.69	-0.11	1.83	1.82	0.64

Table 7: **Performance of Transferring using Mispricing or Risk Characteristics.** This table reports the performance of long-short portfolios constructed using 2stage models where the local component is fitted using either mispricing, risk, or agnostic characteristics. The classification of characteristics follows Chen et al. (2022). All numbers are annualized and in percent. Panel A includes all stocks, while Panel B includes only stocks outside the US.

"Fundamental", and "Miscellaneous" mispricing sub-groups. These contain 18, 19, and 19 characteristics, respectively, so they each have fewer characteristics than the risk group.[11]

Table 8 shows the performance of the local-and-global models that allow local variation from one of these mispricing sub-groups. Even though each group has fewer characteristics to learn a local component from, they all have higher alphas to the global model than the risk-based equivalent. Further, one of these alphas is statistically significant at a 5% level and most are significant at the 10% level. These consistently positive alphas are further evidence of local elements in mispricing. This evidence also suggests that the risk-based characteristics' failure to produce useful local components does not stem from having too few characteristics.

In short, we find risk pricing to be purely global while mispricing has a large global component and a smaller local component. The local variation in mispricing can stem from

[11] "Investment" consists of mispricing factors from the clusters in Jensen et al. (2023) called accruals and investment; "Fundamental" consists of those from the clusters called profit-growth, profitability, quality, size, and value; "Miscellaneous" consists of the remaining clusters. See Appendix F for details.

Panel A: All Stocks

	Mispricing Inv			Mispricing Fund			Mispricing Misc		
α	9.41***	9.29***	0.09	9.36***	9.25***	0.32*	9.60***	9.51***	0.45*
	(6.63)	(6.57)	(1.31)	(6.76)	(6.69)	(1.77)	(6.81)	(6.74)	(1.83)
Mkt		0.03*	0.00		0.03*	0.00		0.02	-0.01**
		(1.71)	(1.01)		(1.70)	(0.38)		(1.29)	(-2.19)
Global			0.99***			0.96***			0.97***
			(370.22)			(134.74)			(98.00)
IR	1.41	1.4	0.3	1.44	1.43	0.41	1.45	1.44	0.42

Panel B: Ex-US Stocks

	Mispricing Inv			Mispricing Fund			Mispricing Misc		
α	9.37***	9.31***	0.18*	9.23***	9.17***	0.61**	9.65***	9.62***	0.79*
	(8.16)	(8.11)	(1.78)	(8.36)	(8.31)	(2.14)	(8.30)	(8.26)	(1.94)
Mkt		0.02	0.00		0.02	0.00		0.01	-0.01
		(1.35)	(1.13)		(1.31)	(0.38)		(0.69)	(-1.61)
Global			0.97***			0.91***			0.94***
			(199.15)			(68.57)			(48.98)
IR	1.74	1.73	0.42	1.78	1.78	0.51	1.77	1.76	0.46

Table 8: **Performance of Transferring using Different Mispricing Characteristics.** This table reports the performance of long-short portfolios constructed using 2stage models where the local component is fitted using different mispricing characteristics. The classification of characteristics follows Chen et al. (2022). Mispricing characteristics are partitioned using clusters from Jensen et al. (2023). All numbers are annualized and in percent. Panel A includes all stocks, while Panel B includes only stocks outside the US.

differences in cultural biases and preferences, institutional settings such as pension fund investment practices, or market structures.

5.6 Estimating Non-Linear Transfer Learning

So far, we have considered linear asset pricing models, which have the benefit of being easily interpretable and consistent with much of the asset pricing literature. Considering non-linear models is also interesting, however, for several reasons. First, non-linear machine learning models are known for their predictive power, so it is interesting to see how far we get by combining this predictive power with that of transfer learning. Second, it is interesting to consider how robust our results are to methodological choices, including the magnitude of the global element analyzed in the next section.

We consider two non-linear ML prediction methods: XGBoost (XGB, see Friedman, 2001; Chen et al., 2015) and random Fourier features (RFF, Rahimi and Recht, 2007, 2008). Based on each of these two methods, we have created local, hard, global, soft transfer, and 2stage predictions. Specifically, the single-country methods (local, hard, global) are estimated using XGB and RFF, respectively.

The two-step methods (soft and 2stage) are estimated based on the method presented in Section 3.3. In the first step, we estimate the US or global model using either XGB or RFF.[12] In the second step, we estimate additional local components as a linear model estimated using an elastic net.

In other words, for the local-and-global models, we use different methods to estimate the global and local components, which is feasible within the framework of Section 3.3. The choice of methods is motivated by the following considerations. In the first step, we use a large global data set, so applying non-linear machine-learning methods is promising. In the second step, we estimate the local effects using a simple linear function due to the limited amount of data in the individual countries.

The performance of these non-linear prediction procedures is summarized in Tables 9 and 10. Panel A of Table 9 presents the predictive performance as measured by R^2_{OOS} of the five transfer settings (local, hard, global, soft transfer, and 2stage) based on, respectively, RFF and XGB. Similarly, Panel B of Table 9 shows the Sharpe ratio of each long-short factor based on the predictions of the different models.

The table shows a poor performance of the purely local models, both in terms of R^2_{OOS} and Sharpe ratio. The XGB method performs particularly poorly for the local models, with a negative R^2_{OOS} and a Sharpe ratio below that of the corresponding linear model, likely because of the more complex and flexible nature of XGB, which does not work well with a small data set.

In contrast, the non-linear models perform well for the purely global models (hard and global) and the local-and-global transfer learning models (soft and 2stage). For example, these methods deliver Sharpe ratios above 2, whereas the linear models had Sharpe ratios below 2.

In Table 10, we further investigate the performance of the factors constructed using the local-and-global models (soft and 2stage) based on RFF and XGB. We see the same

[12] In all applications of RFF, we have standardized all random features country-cross-sectionally. Hyperparameters are tuned using the same techniques outlined in Section 4.2.

Panel A: R^2_{OOS}

	Local	Hard	Global	Soft	2Stage
RFF	0.07	0.205	0.295	0.313	0.362
XGB	-0.15	0.585	0.669	0.560	0.683

Panel B: Sharpe Ratio

RFF	0.844	2.30	2.39	2.41	2.43
XGB	0.485	1.96	2.11	2.14	2.19

Table 9: **Performance of Non-linear Transfer Learning.** This table reports the performance of the local, hard, global, soft, and 2stage models, where the estimation functions are achieved using ML methods. For soft transfer and 2stage, the procedure described in Section 3.3 is applied using an ML method, either RFF or XGB, for finding $\hat{f}$ and a linear model for finding $\tilde{g}$. Panel A shows the predictive performance measured by R^2_{OOS}. Panel B shows the Sharpe ratio of a factor aggregated from country-wise long-short factors based on each model's performance.

overall pattern as we have previously observed for the similar linear models in Panel A of Table 4: Both soft and 2stage, in each of the cases RFF and XGB, exhibit significant average returns, CAPM alphas, and alphas to the global and local factors (constructed using the corresponding ML method).

These results illustrate that our transfer learning models can be applied with different non-linear ML methods, even when combining different ML methods in each step of the estimation. Further, the result reflect a strong global component and a smaller local one, as discussed in more detail next.

5.7 How Global is Predictability? A New Ratio

To estimate the relative magnitudes of the global and local effects, we define a measure of the relative importance of the global element for any country c as

$$\mathcal{G}_c = \frac{||\theta_g||_2}{||\theta_g||_2 + ||\theta_{\ell,c}||_2}. \tag{28}$$

Panel A: RFF

	Soft				2Stage			
α	10.76***	10.75***	1.31**	1.15**	10.87***	10.85***	0.59*	0.81**
	(11.28)	(11.25)	(2.41)	(2.11)	(11.38)	(11.34)	(1.81)	(2.58)
Mkt		0.00	-0.01**	-0.01*		0.01	-0.01***	-0.01***
		(0.30)	(-2.04)	(-1.81)		(0.54)	(-2.94)	(-3.61)
Global			0.90***	0.94***			0.98***	0.93***
			(30.66)	(26.39)			(55.93)	(45.13)
Local				-0.07*				0.10***
				(-1.88)				(4.77)
IR	2.41	2.4	0.63	0.56	2.43	2.42	0.47	0.68

Panel B: XGBoost

	Soft				2Stage			
α	11.79***	11.88***	1.72***	1.25*	12.40***	12.44***	0.86**	1.12***
	(10.03)	(10.15)	(2.62)	(1.93)	(10.27)	(10.29)	(2.44)	(3.21)
Mkt		-0.02**	-0.02***	-0.01***		-0.01	-0.00	-0.00
		(-2.00)	(-2.99)	(-2.61)		(-0.88)	(-1.00)	(-1.48)
Global			0.84***	0.91***			0.96***	0.92***
			(29.71)	(28.14)			(63.04)	(53.18)
Local				-0.12***				0.07***
				(-4.01)				(4.12)
IR	2.14	2.17	0.66	0.49	2.19	2.2	0.61	0.82

Table 10: **Performance of Non-linear Transfer Learning.** This table reports the performance of long-short portfolios constructed using global, soft, and 2stage models, where the estimation functions are achieved using ML methods. For soft transfer and 2stage, the procedure described in Section 3.3 is applied using an ML method for finding $\hat{f}$ and a linear model for finding $\tilde{g}$. In Panel A the ML method is RFF, and in Panel B the method is XGB.

In other words, the "global element" $\mathcal{G}_c$ is the relative magnitude of the global parameters, and the corresponding "local element" is $1 - \mathcal{G}_c$. Averaging across countries, we compute the overall global importance as

$$\mathcal{G} = \frac{1}{C} \sum_{c=1}^{C} \mathcal{G}_c. \tag{29}$$

To estimate the relative global element $\mathcal{G}$ empirically, we use our joint model. The joint model is useful because it immediately shows what the local and global components are, where this separation is less direct for our other models due to the transfer parameter ν (as discussed in Section 5.4).

For each country, we compute the posterior distribution of θ_g and $\theta_{\ell,c}$ given the observed data, as explained in Appendix C.3. We then draw 100,000 random samples of $(\theta_g, \theta_{\ell,1}, ..., \theta_{\ell,C})$ from the posterior. For each random sample, we then compute $\mathcal{G}_c$ and $\mathcal{G}$, and then take the average as our estimate of the relative global components.

Figure 6 shows each country's estimated $\mathcal{G}_c$ as well as the overall global importance $\mathcal{G}$. We see that the overall relative global component $\mathcal{G}$ is estimated to be 96%, a very large fraction. In addition, $\mathcal{G}_c$ is estimated to be high in all countries, although with some variation. The countries are sorted by the amount of data available in the training period. The figure does not reveal a strong connection between the global importance and the amount of data.[13]

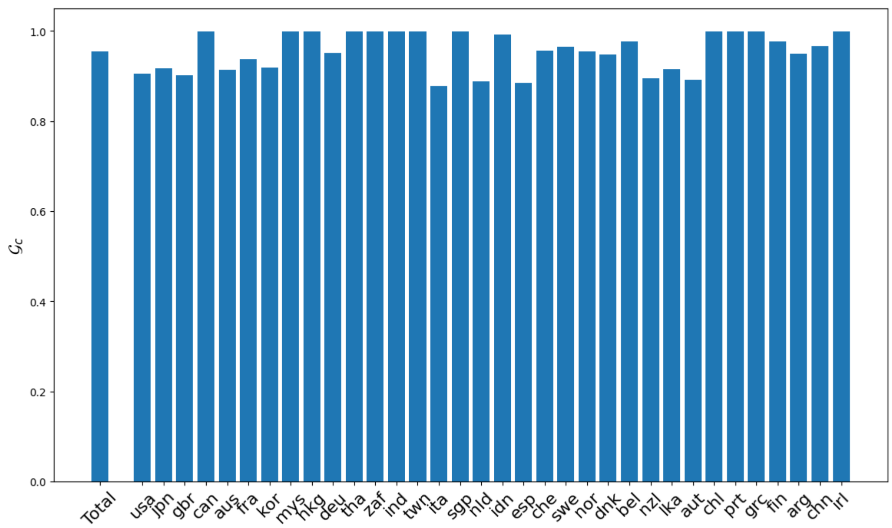

Figure 6: **Relative Global Importance.** This figure plots, for each country, the importance of the global component for predicting local stock returns as defined in (28). Further, 'Total' shows the average global importance as defined in (28), which is 96%. Hyperparameters for joint model are tuned using data up until 1999, and those hyperparameters are then used to fit the joint model on a sample of global stocks 2000-2021.

The time variation of the globalness is also interesting, as illustrated in Figure 7. The figure plots the global importance over time using a 5-year rolling window. As previously, the hyperparameters for the joint model are tuned using data up until 1999, but now the test data is a rolling 5 year subset of the original test data. We see that the globalness is generally high and stable across our sample.

As mentioned earlier, globalness is not easily defined for our linear and non-linear 2stage models based on (29). However, in these cases, we can compute another measure of global importance. In particular, we compute the correlation between the *predictions* of the purely global model with those of the 2stage model for each country, and then average these across

[13] In Appendix D, we further investigate the connection between the number of training observations in a country and its globalness, finding no significant relation.

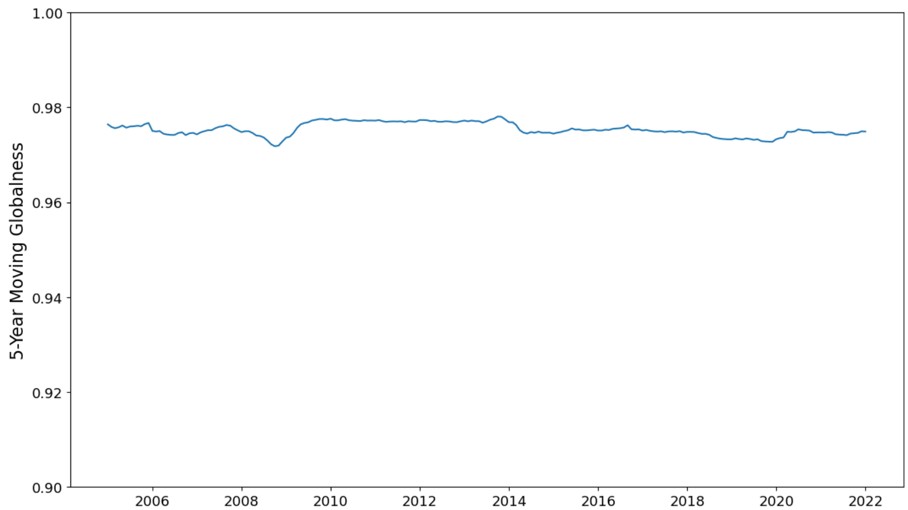

Figure 7: **5-Year Rolling Globalness** This figure plots the 5-year rolling average globalness. Hyperparameters for the joint model are tuned using data up until 1999, and those hyperparameters are then used to fit the joint model on a sample of global stocks 2000-2021.

countries.

Figure 8 shows the resulting overall global importance for the linear model and non-linear models based on RFF and XGB. The global importance is 92% for the linear model, 95% for RFF, and 96% for XGB. In other words, we again find high levels of global importance across methods, comparable to our estimate from the joint linear model based on (29).

In sum, we estimate that the global component dominates the predictive parameter, with a relative global component north of 90%. This result is consistent with our findings based on R-squared and portfolio performance, namely that the global model vastly outperforms the local one, and the local-and-global only slightly outperforms the purely global model. This result is robust to globalness measure and model types.

6 Conclusion

We find that expected returns are almost the same function of stock characteristics in all countries and the function appears to have been relatively stable over the past century. These predictive coefficients are an essential building block of the stochastic discount factor, so our results imply that the stochastic discount factor is a function of stock characteristics that is similar across countries and time.

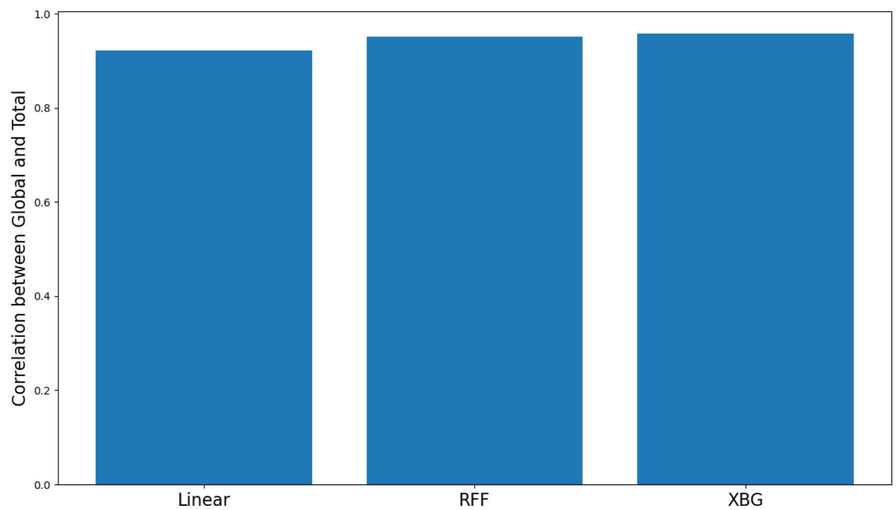

Figure 8: **Global Importance: Linear and Non-Linear Models.** This figure plots the cross-country average correlation between the predictions of the purely global model and the predictions of 2stage for linear and non-linear models. The cross-country average correlations are 92%, 95%, and 96%, respectively. Models are tuned and trained on data up until 1999, and their predictions are out-of-sample predictions on a sample of global stocks 2000-2021.

Further, our results imply that transferring information about how characteristics predict returns across countries and regions can significantly improve return predictions. Indeed, using global information materially improves local models in many countries and can even be helpful in a large country such as the US.

While the common global component of return prediction is estimated to be 96%, small, but significant, local components can be identified using the transfer learning method that we develop. We show theoretically and empirically that this method lowers expected prediction errors. Our method sheds new light on what characteristics predict returns "everywhere" and what the local differences are. We find local differences are driven by mispricing-related signals rather than risk-based ones.

From an asset pricing perspective, the paper raises a number of interesting questions for future research: Why are expected returns so similar across countries? To what extent is the global component driven by global risk pricing, universal human behavioral instincts, common market structures, or market integration? Is the local component due to differences in culture, institutions, or market structures?

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ONLINE APPENDIX

The appendix is organized as follows. Appendix A shows the closed-form solution and behavior of the GENet when only using ridge penalization. Appendix B presents the proofs of all our propositions. Appendix C explains the Bayesian interpretations of the GENet and joint methods. Appendix D studies the relevance of the number of training observations across countries. Appendix E shows the spaces we tune hyperparameters over for all models considered. Appendix F contains the table of Jensen et al. (2023) characteristisc used and their risk- or mispricing-classification.

A GENet and transfer learning in the ridge case

In the ridge case, the GENet can be solved in a closed form, as seen in the following lemma. The solution provides additional information on the potential benefits of the method and, subsequently, leads to closed form expressions for both the single-region estimators introduced in Section 3.1 and the two-step learning estimators from Section 3.2.

Lemma 1 *The estimator $\hat{\theta}(\hat{\theta}_0) = \hat{\theta}^{GENet}_{c,\nu,\lambda,\alpha=0}(\hat{\theta}_0)$ from (10) in the ridge case (i.e., $\alpha = 0$) is given by*

$$\hat{\theta}(\hat{\theta}_0) = R(X_c' Y_c + \lambda\nu\hat{\theta}_0), \tag{A.1}$$

where $R = (X_c' X_c + \lambda I_p)^{-1}$. The estimator's bias is

$$\mathbb{E}[\hat{\theta}(\hat{\theta}_0) - \theta | \theta, \hat{\theta}_0] = \lambda R(\nu\hat{\theta}_0 - \theta), \tag{A.2}$$

and its variance is

$$\mathrm{Var}_\theta[\hat{\theta}(\hat{\theta}_0) | \theta, \hat{\theta}_0] = R X_c' \mathrm{Var}[Y_c] X_c R. \tag{A.3}$$

Proof. In the ridge case, the minimization problem simplifies to

$$\hat{\theta}(\hat{\theta}_0) = \underset{\theta\in\mathbb{R}^p}{\arg\min}\{(Y_c - X_c\theta)'(Y_c - X_c\theta) + \lambda\|\theta - \nu\hat{\theta}_0\|_2^2,$$

which is differentiable. Taking derivatives and setting it equal to zero gives the expression for $\hat{\theta}(\hat{\theta}_0)$. Using this, it is straightforward to show

$$\mathbb{E}[\hat{\theta}|\hat{\theta}_0] - \theta = RX_c'\mathbb{E}[Y] + \lambda R\nu\hat{\theta}_0 - \theta = \lambda R(\nu\hat{\theta}_0 - \theta).$$

Finally, we find similarly that

$$\mathrm{Var}[\hat{\theta}|\hat{\theta}_0] = R\mathrm{Var}[X_c'Y_c + \lambda\nu\hat{\theta}_0|\hat{\theta}_0]R = RX_c'\mathrm{Var}[Y_c]X_cR.$$

□

The first part of the proposition shows the closed-form solution of the estimator. The second part of the proposition shows that the estimator has a smaller bias when the scaled initial guess, $\nu\theta_0$, is closer to the true parameter, θ. The third part of the proposition shows that the estimator's variance is independent of the guess θ_0.

A central insight in machine learning is that we can make better predictions by controlling the bias plus the variance. Since Lemma 1 shows that the variance is independent of θ_0 and the bias decreases in $\nu\theta_0 - \theta$, we see that $\hat{\theta}(\theta_0)$ is superior to the regular ridge estimator, $\hat{\theta}(0)$, as long as the initial guess has a smaller distance to the truth $\nu\theta_0 - \theta$ than the distance between zero (no guess) and the truth.

Clearly, when θ_0 is a good guess that is known without error, choosing $\nu = 1$ minimizes the bias at no cost in terms of variance, as seen from Lemma 1. However, in the more realistic case when θ_0 is estimated, then estimation noise contributes to the variance, so the bias-variance trade-off can result in an optimal ν less than one, as discussed more precisely in Section 3.4.

As a simple consequence of Lemma 1 we have closed form expressions of the single-region estimators from Section 3.1 and the two-step learning estimators from Section 3.2. In the notation for the estimators we highlight the dependence of all hyperparameters.

Lemma 2 *When estimation is performed using ridge estimation, our estimators from Sections 3.1 and 3.2 take the following form*

$$
\begin{aligned}
&\textit{local} && \hat{\theta}^{local}_{c,\lambda} = \hat{\theta}^{ENet}_{c,\lambda,0} = R_{c,\lambda} X_c' Y_c \\
&\textit{hard} && \hat{\theta}^{hard}_{c,\lambda_0} = \hat{\theta}^{ENet}_{c_0,\lambda_0,0} = R_{c_0,\lambda_0} X_{c_0}' Y_{c_0} \\
&\textit{global} && \hat{\theta}^{global}_{c,\lambda_0} = \hat{\theta}^{ENet}_{\mathcal{C},\lambda_0,0} = R_{\mathcal{C},\lambda} X_{\mathcal{C}}' Y_{\mathcal{C}} \\
&\textit{soft transfer} && \hat{\theta}^{soft}_{c,\nu,\lambda,\lambda_0} = \hat{\theta}^{GENet}_{c,\nu,\lambda,0}(\hat{\theta}^{hard}_{c,\lambda_0}) = R_{c,\lambda} X_c' Y_c + \nu\lambda R_{c,\lambda} R_{c_0,\lambda_0} X_{c_0}' Y_{c_0} \\
&\textit{two stage} && \hat{\theta}^{2stage}_{c,\nu,\lambda,\lambda_0} = \hat{\theta}^{GENet}_{c,\nu,\lambda,0}(\hat{\theta}^{global}_{c,\lambda_0}) = R_{c,\lambda} X_c' Y_c + \nu\lambda R_{c,\lambda} R_{\mathcal{C},\lambda_0} X_{\mathcal{C}}' Y_{\mathcal{C}}
\end{aligned}
\tag{A.4}
$$

Proof. This follows from straightforward manipulations using Lemma 1. □

B Proofs

Proof of Proposition 1(i). For the estimator $\hat{\theta}^{\text{soft}}_{c,\nu,\lambda,\lambda_0}$ we find, making straightforward manipulations from (A.4), that

$$
\begin{aligned}
MSE(\hat{\theta}^{\text{soft}}_{c,\nu,\lambda,\lambda_0}) =& MSE(\hat{\theta}^{\text{local}}_{c,\lambda}) + \nu^2\lambda^2 \mathbb{E}\|R_{c,\lambda}\hat{\theta}^{\text{local}}_{c_0,\lambda_0}(0)\|_2^2 \\
&+ 2\nu\lambda\mathbb{E}\big((\hat{\theta}^{\text{local}}_{c,\lambda}(0) - \theta_c)' R_{c,\lambda}\hat{\theta}^{\text{local}}_{c_0,\lambda_0}(0)\big)
\end{aligned}
$$

The statement follows from differentiating with respect to ν, if we can argue that $\mathbb{E}\big((\hat{\theta}^{\text{local}}_{c,\lambda} - \theta_c)' R_{c,\lambda}\hat{\theta}^{\text{local}}_{c_0,\lambda_0}\big) < 0$. For this we first rewrite the conditional expectation using the conditional independence between Y_c and Y_{c_0},

$$
\begin{aligned}
&\mathbb{E}\big((\hat{\theta}^{\text{local}}_{c,\lambda}(0) - \theta_c)' R_{c,\lambda}\hat{\theta}^{\text{local}}_{c_0,\lambda_0}(0) \mid \Theta\big) \\
&= \big(R_{c,\lambda} X_c' X_c \theta_c - \theta_c\big)' R_{c,\lambda} R_{c_0,\lambda_0} X_{c_0}' X_{c_0} \theta_{c_0} \\
&= -\big((I_p - R_{c,\lambda} X_c' X_c)(\theta_g + \theta_{\ell,c})\big)' R_{c,\lambda} R_{c_0,\lambda_0} X_{c_0}' X_{c_0} (\theta_g + \theta_{\ell,c_0}) \\
&= -(\theta_g + \theta_{\ell,c})' \lambda R^2_{c,\lambda} R_{c_0,\lambda_0} X_{c_0}' X_{c_0} (\theta_g + \theta_{\ell,c_0}).
\end{aligned}
$$

Taking expectations and utilizing the joint independence of $\theta_g, \theta_{\ell,c}$ and θ_{ℓ,c_0} then gives

$$\begin{aligned}
&\mathbb{E}\big((\hat{\theta}^{\text{local}}_{c,\lambda} - \theta_c)' R_{c,\lambda} \hat{\theta}_{c_0,\lambda_0}\big) \\
&= -\mathbb{E}\big(\theta_g' \lambda R^2_{c,\lambda} R_{c_0,\lambda_0} X'_{c_0} X_{c_0} \theta_g\big) \\
&= -\lambda \sigma_g^2 \operatorname{tr}(R^2_{c,\lambda} R_{c_0,\lambda_0} X'_{c_0} X_{c_0}) \\
&= -\lambda \sigma_g^2 \operatorname{tr}(R_{c,\lambda}(I_p + \lambda_0 (X'_{c_0} X_{c_0})^{-1})^{-1} R_{c,\lambda}),
\end{aligned}$$

which is strictly negative, since the matrix inside the trace-function is seen to be positive definite. □

The proof of Proposition 1(ii)-(iii) relies on Lemma 3 below. We formulate this as a separate result, since it demonstrates that having data from many countries makes the first step in the two stage estimator close to the optimal reference point, the true global parameter θ_g. We give the proof of this lemma, before we proceed with the proof of Proposition 1.

Lemma 3 *Under Assumptions 1–3 it holds that* $\hat{\theta}^{global}_{c,\lambda_0} \to \theta_g$ *and* $\hat{\theta}^{2stage}_{c,\nu,\lambda,\lambda_0} \to \hat{\theta}^{GENet}_{c,\nu,\lambda,0}(\theta_g)$ *as* $C \to \infty$*, where these convergences hold in* L^2 *and almost surely.*[14]

Proof of Lemma 3. We rewrite

$$\hat{\theta}^{\text{global}}_{c,\lambda_0} = R_{\mathcal{C},\lambda_0} X'_{\mathcal{C}} Y_{\mathcal{C}} \tag{B.1}$$

$$= \big(\frac{1}{C}\sum_c X'_c X_c + \frac{\lambda_0}{C} I_p\big)^{-1}\big(\frac{1}{C}\sum_c X'_c X_c\big)\theta_g \tag{B.2}$$

$$+ \big(\frac{1}{C}\sum_c X'_c X_c + \frac{\lambda_0}{C} I_p\big)^{-1}\big(\frac{1}{C}\sum_c X'_c X_c \theta_{\ell,c}\big) \tag{B.3}$$

$$+ \big(\frac{1}{C}\sum_c X'_c X_c + \frac{\lambda_0}{C} I_p\big)^{-1}\big(\frac{1}{C}\sum_c X'_c \varepsilon_c\big) \tag{B.4}$$

For the convergence results on $\hat{\theta}^{\text{global}}_{c_1,\lambda_0}$ it suffices to demonstrate almost sure and L^2 convergence for the three terms in (B.2)-(B.4) separately. The terms will have limits θ_g, 0 and 0.

[14] Convergence in L^2 means that, e.g., $\mathbb{E}(\|\hat{\theta}^{\text{global}}_c - \theta_g\|_2^2) \to 0$, where all variables are square integrable.

The almost sure convergence follows from the strong law of large numbers, since all expectations $\mathbb{E}(X_c'X_c)$, $\mathbb{E}(X_c'X_c\theta_{\ell,c})$ and $\mathbb{E}(X_c'\varepsilon_c)$ are well defined with the two latter being zero due to the assumptions on integrability of $X_c'X_c$ and relevant (conditional) independence between $X_c\theta_g, \theta, \theta_{\ell,c}$, where $\theta_{\ell,c}$ and ε_c have mean zero. Also note that $\mathbb{E}(X_c'X_c)$ is positive definite, so the inverse exists (and is positive definite).

To demonstrate the L^2-convergence of (B.2) we write

$$\begin{aligned}
&\mathbb{E}\Big(\|(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-1}(\frac{1}{C}\sum_c X_c'X_c)\theta_g - \theta_g\|_2^2\Big)\\
&= \mathbb{E}\Big(\|\frac{\lambda_0}{C}(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-1}\theta_g\|_2^2\Big)\\
&= \mathbb{E}\Big(\theta_g'\frac{\lambda_0^2}{C^2}(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-2}\theta_g\Big)\\
&= \sigma_g^2\mathbb{E}\Big(\mathrm{tr}(\frac{\lambda_0^2}{C^2}(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-2})\Big).
\end{aligned}$$

This converges to zero due to the strong law of large number and dominated convergence, using that

$$\frac{\lambda_0}{C}(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-1} \leq I_p,$$

where inequality means that the difference is positive semi-definite. Similarly, we demonstrate L^2-convergence of (B.3) by writing

$$\begin{aligned}
&\mathbb{E}\Big(\|(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-1}(\frac{1}{C}\sum_c X_c'X_c\theta_{\ell,c})\|_2^2\Big)\\
&= \mathbb{E}\Big((\frac{1}{C}\sum_c \theta_{\ell,c}'X_c'X_c)(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-2}(\frac{1}{C}\sum_c X_c'X_c\theta_{\ell,c})\Big)\\
&= \sigma_\ell^2\mathbb{E}\Big(\mathrm{tr}((\frac{1}{C^2}\sum_c (X_c'X_c)^2)(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p)^{-2})\Big),
\end{aligned}$$

which has limit zero due to the law of large numbers, using integrability of $(X_c'X_c)^2$, and dominated convergence. In the application of dominated convergence we use that

$$\sum_c (X_c'X_c)^2 \leq \big(\sum_c (X_c'X_c)\big)^2.$$

For the L^2-convergence of (B.4) we have

$$\begin{aligned}
&\mathbb{E}\Big(\big\|\big(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p\big)^{-1}\big(\frac{1}{C}\sum_c X_c'\varepsilon_c\big)\big\|_2^2\Big)\\
&= \mathbb{E}\Big(\big(\frac{1}{C}\sum_c \varepsilon_c'X_c\big)\big(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p\big)^{-2}\big(\frac{1}{C}\sum_c X_c'\varepsilon_c\big)\Big)\\
&= \sigma_\varepsilon^2\mathbb{E}\Big(\mathrm{tr}\big(\big(\frac{1}{C^2}\sum_c X_c'X_c\big)\big(\frac{1}{C}\sum_c X_c'X_c + \frac{\lambda_0}{C}I_p\big)^{-2}\big)\Big),
\end{aligned}$$

which with similar arguments is seen to converge to 0.

For the statement on $\hat{\theta}^{\text{2stage}}_{c,\nu,\lambda,\lambda_0}$, we recall that

$$\hat{\theta}^{\text{2stage}}_{c,\nu,\lambda,\lambda_0} = R_{c,\lambda}X_c'Y_c + \nu\lambda R_{c,\lambda}\hat{\theta}^{\text{global}}_{c,\lambda_0},$$

which, due to the equivalent convergence of $\hat{\theta}^{\text{global}}_{c,\lambda_0}$, converges both almost surely and in L^2 to

$$R_{c,\lambda}X_c'Y_c + \nu\lambda R_{c,\lambda}\theta_g$$

To conclude the L^2-convergence, we use that $\lambda R_{c,\lambda} \leq I_p$. $\square$

Proof of Proposition 1(ii)-(iii). It follows directly from Lemma 3 that

$$MSE(\hat{\theta}^{\text{2stage}}_{c,\nu,\lambda,\lambda_0}) \to MSE(\hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,0}(\theta_g))$$

as $C \to \infty$. We use that $\hat{\theta}^{\text{GENet}}_{c,\nu,\lambda,0}(\theta_g) = \hat{\theta}^{\text{local}}_{c,\lambda} + \nu\lambda R_{c,\lambda}\theta_g$ to rewrite

$$
\begin{aligned}
&MSE(\hat{\theta}^{\text{local}}_{c,\lambda}(\nu\theta_g)) \\
&= MSE(\hat{\theta}^{\text{local}}_{c,\lambda}) + \nu^2\lambda^2\mathbb{E}\big(\|R_{c,\lambda}\theta_g\|_2^2\big) + 2\nu\lambda\mathbb{E}\big(\theta_g' R_{c,\lambda}(R_{c,\lambda}X_c'Y_c - \theta_c)\big) \\
&= MSE(\hat{\theta}^{\text{local}}_{c,\lambda}) + \nu^2\lambda^2\mathbb{E}\big(\theta_g' R_{c,\lambda}^2\theta_g\big) + 2\nu\lambda\mathbb{E}\big(\theta_g' R_{c,\lambda}(R_{c,\lambda}X_c'X_c - I_p)\theta_g\big) \\
&= MSE(\hat{\theta}^{\text{local}}_{c,\lambda}) + \nu^2\lambda^2\sigma_g^2\mathbb{E}\big(\text{tr}(R_{c,\lambda}^2)\big) + 2\nu\lambda\sigma_g^2\mathbb{E}\big(\text{tr}(R_{c,\lambda}(R_{c,\lambda}X_c'X_c - I_p))\big) \\
&= MSE(\hat{\theta}^{\text{local}}_{c,\lambda}) + \lambda^2\nu(\nu-2)\sigma_g^2\mathbb{E}\big(\text{tr}(R_{c,\lambda}^2)\big),
\end{aligned}
$$

from which the statements (ii) and (iii) of the proposition follows. □

Proof of Proposition 2. For the statements on the soft transfer estimator, we write

$$
\begin{aligned}
MSE(\hat{\theta}^{\text{soft}}_{c,1,\lambda,\lambda_0}) =& MSE(\hat{\theta}^{\text{hard}}_{c_0,\lambda_0}) + \mathbb{E}\|(\lambda R_{c,\lambda} - I_p)R_{c_0,\lambda_0}X_{c_0}'Y_{c_0} + R_{c,\lambda}X_c'Y_c\|_2^2 \\
&+ 2\mathbb{E}\big((R_{c_0,\lambda_0}X_{c_0}'Y_{c_0} - \theta_c)'(\lambda R_{c,\lambda} - I_p)R_{c_0,\lambda_0}X_{c_0}'Y_{c_0}\big) \\
&+ 2\mathbb{E}\big((R_{c_0,\lambda_0}X_{c_0}'Y_{c_0} - \theta_c)'R_{c,\lambda}X_c'Y_c\big).
\end{aligned}
\tag{B.5}
$$

We consider each of the three expectations separately. The first can – using the same conditioning techniques as in the proofs of Proposition 1 and Lemma 3 – be calculated as

$$
\begin{aligned}
&\mathbb{E}\|(\lambda R_{c,\lambda} - I_p)R_{c_0,\lambda_0}X_{c_0}'Y_{c_0} + R_{c,\lambda}X_c'Y_c\|_2^2 \\
&= \mathbb{E}\big(Y_{c_0}'X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda} - I_p)^2R_{c_0,\lambda_0}X_{c_0}'Y_{c_0}\big) \\
&\quad + \mathbb{E}\big(Y_c'X_cR_{c,\lambda}^2X_c'Y_c\big) \\
&\quad + 2\mathbb{E}(Y_c'X_cR_{c,\lambda}(\lambda R_{c,\lambda} - I_p)R_{c_0,\lambda_0}X_{c_0}'Y_{c_0}\big) \\
&= \sigma_\varepsilon^2\text{tr}(X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda} - I_p)^2R_{c_0,\lambda_0}X_{c_0}') \\
&\quad + (\sigma_g^2 + \sigma_\ell^2)\text{tr}(X_{c_0}'X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda} - I_p)^2R_{c_0,\lambda_0}X_{c_0}'X_{c_0}) \\
&\quad + \sigma_\varepsilon^2\text{tr}(X_cR_{c,\lambda}^2X_c') \\
&\quad + (\sigma_g^2 + \sigma_\ell^2)\text{tr}(X_c'X_cR_{c,\lambda}^2X_c'X_c) \\
&\quad + 2\sigma_g^2\text{tr}(X_c'X_cR_{c,\lambda}(\lambda R_{c,\lambda} - I_p)R_{c_0,\lambda_0}X_{c_0}'X_{c_0}).
\end{aligned}
\tag{B.6}
$$

For the second expectation in (B.10) we have

$$
\begin{aligned}
&\mathbb{E}\big((R_{c_0,\lambda_0}X'_{c_0}Y_{c_0}-\theta_c)'(\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0}Y_{c_0}\big)\\
&=\mathbb{E}\big(Y'_{c_0}X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0}Y_{c_0}\big)\\
&\quad-\mathbb{E}\big(\theta'_c(\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0}Y_{c_0}\big)\\
&=\sigma_\varepsilon^2\mathrm{tr}(X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0})\\
&\quad+(\sigma_g^2+\sigma_\ell^2)\mathrm{tr}(X'_{c_0}X_{c_0}R_{c_0,\lambda_0}(\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0}X_{c_0})\\
&\quad-\sigma_g^2\mathrm{tr}((\lambda R_{c,\lambda}-I_p)R_{c_0,\lambda_0}X'_{c_0}X_{c_0}).
\end{aligned}
\tag{B.7}
$$

Finally, for the third expectation in (B.10) we find

$$
\begin{aligned}
&\mathbb{E}\big((R_{c_0,\lambda_0}X'_{c_0}Y_{c_0}-\theta_c)'R_{c,\lambda}X'_cY_c\big)\\
&=\sigma_g^2\mathrm{tr}(X'_{c_0}X_{c_0}R_{c_0,\lambda_0}R_{c,\lambda}X'_cX_c)\\
&\quad-\sigma_g^2\mathrm{tr}(R_{c,\lambda}X'_cX_c)\\
&\quad-\sigma_\ell^2\mathrm{tr}(R_{c,\lambda}X'_cX_c).
\end{aligned}
\tag{B.8}
$$

Now, using that

$$
\begin{aligned}
&\lim_{\lambda\to\infty}\lambda R_{c,\lambda}=I_p,\\
&\lim_{\lambda\to\infty}\lambda(\lambda R_{c,\lambda}-I_p)=-X'_cX_c,
\end{aligned}
$$

we find

$$
\begin{aligned}
&\lim_{\lambda\to\infty}\lambda\big(MSE(\hat\theta^{\text{soft}}_{c,1,\lambda,\lambda_0})-MSE(\hat\theta^{\text{hard}}_{c_0,\lambda_0})\big)\\
&=-2\sigma_\varepsilon^2\mathrm{tr}(X_{c_0}R_{c_0,\lambda_0}X'_cX_cR_{c_0,\lambda_0}X'_{c_0})\\
&\quad-2(\sigma_g^2+\sigma_\ell^2)\mathrm{tr}(X'_{c_0}X_{c_0}R_{c_0,\lambda_0}X'_cX_cR_{c_0,\lambda_0}X'_{c_0}X_{c_0})\\
&\quad+4\sigma_g^2\mathrm{tr}(X'_cX_cR_{c_0,\lambda_0}X'_{c_0}X_{c_0})\\
&\quad-2\sigma_g^2\mathrm{tr}(X'_cX_c)\\
&\quad-2\sigma_\ell^2\mathrm{tr}(X'_cX_c).
\end{aligned}
\tag{B.9}
$$

Now the sum of all terms involving σ_g^2 will tend to 0 as $\lambda_0 \to 0$, and thereby the limit will be strictly negative for λ_0 small enough.

For the statements on the two stage estimator we rewrite, similarly to (B.10),

$$
\begin{aligned}
MSE(\hat{\theta}^{\text{2stage}}_{c,1,\lambda,\lambda_0}) =& MSE(\hat{\theta}^{\text{global}}_{c,\lambda_0}) + \mathbb{E}\|(\lambda R_{c,\lambda} - I_p)R_{\mathcal{C},\lambda_0}X_{\mathcal{C}}'Y_{\mathcal{C}} + R_{c,\lambda}X_c'Y_c\|_2^2 \\
&+ 2\mathbb{E}\big((R_{\mathcal{C},\lambda_0}X_{\mathcal{C}}'Y_{\mathcal{C}} - \theta_c)'(\nu\lambda R_{c,\lambda} - I_p)R_{\mathcal{C},\lambda_0}X_{\mathcal{C}}'Y_{\mathcal{C}}\big) \\
&+ 2\mathbb{E}\big((R_{\mathcal{C},\lambda_0}X_{\mathcal{C}}'Y_{\mathcal{C}} - \theta_c)'R_{c,\lambda}X_c'Y_c\big).
\end{aligned}
\tag{B.10}
$$

The result follows from arguments of the same type as for the soft transfer estimator. ☐

Proof of Proposition 3. The minimization problem defining $\hat{\theta}_g, \hat{\theta}_\ell$ can be rewritten as the following:

$$
(\hat{\theta}_g, \hat{\theta}_\ell) = \arg\min_{\theta_g,\theta_\ell} \big\{(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g - Z\theta_\ell)'(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g - Z\theta_\ell) + \lambda_g\theta_g'\theta_g + \theta_\ell'\Lambda\theta_\ell\big\}.
$$

The objective function is strictly convex, so there is a unique global minimum, and to find this it suffices to solve the first order equations. the equations are

$$
X_{\mathcal{C}}'(X_{\mathcal{C}}\theta_g + Z\theta_\ell - Y_{\mathcal{C}}) + \lambda\theta_g = 0 \tag{B.11}
$$

and

$$
Z'(X_{\mathcal{C}}\theta_g + Z\theta_\ell - Y_{\mathcal{C}}) + \Lambda\theta_\ell = 0. \tag{B.12}
$$

Solving (B.11) and (B.12) with respect to θ_g and θ_ℓ, respectively, gives

$$
\theta_g = (X_{\mathcal{C}}'X_{\mathcal{C}} + \lambda I_p)^{-1}X_{\mathcal{C}}'(Y_{\mathcal{C}} - Z\theta_\ell) \tag{B.13}
$$

and

$$
\theta_\ell = (Z'Z + \Lambda)^{-1}(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g). \tag{B.14}
$$

Inserting the expression for θ_ℓ from (B.14) into (B.13) then gives

$$\theta_g = (X'_{\mathcal{C}}X_{\mathcal{C}} + \lambda I_p)^{-1}X'_{\mathcal{C}}\big(Y_{\mathcal{C}} - Z(Z'Z + \Lambda)^{-1}Z'(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g)\big).$$

Solving this with respect to θ_g gives (19), and then (20) follows from (B.14). □

Assumption 5 *(i) Conditional on Θ, $X_{\mathcal{C}}$ and $(\lambda_{\ell,1},\ldots,\lambda_{\ell,C})$, the observations $Y_{\mathcal{C}}$ are generated by (4), where $\varepsilon_1,\ldots,\varepsilon_C$ are independent and identically distributed with $\mathbb{E}(\varepsilon_c) = 0$ and $\mathrm{Var}(\varepsilon_c) = \sigma_\varepsilon^2 I_{N_c}$.*

(ii) Conditioned on $X_{\mathcal{C}}$ and $(\lambda_{\ell,1},\ldots,\lambda_{\ell,C})$, $\theta_g,\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are independent zero-mean parameters, such that $\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are identically distributed, $\mathrm{Var}(\theta_g) = \sigma_g^2 I_p$, $\mathrm{Var}(\theta_{\ell,c}) = \sigma_\ell^2 I_p$, where $\sigma_g^2 > 0$ and $\sigma_\ell^2 \geq 0$.

(iii) The variables $(X_1,\lambda_{\ell,1}),\ldots,(X_C,\lambda_{\ell,C})$ are independent and identically distributed such that X_c has full column rank with probability 1, $\mathbb{E}\|(X_c'X_c)^2\| < \infty$ and $(X_1,\lambda_{\ell,1}),\ldots,(X_C,\lambda_{\ell,C})$ and $\theta_g,\theta_{\ell,1},\ldots,\theta_{\ell,C}$ are jointly independent.

Proof of Proposition 4. We prove the proposition under Assumption 5, which is more general than Assumptions 1–4. The proof follows the lines of the Proof of Proposition 1(ii)-(iii) and Lemma 3. Using that $I_p - X_c'X_c(X_c'X_c + \lambda_c I_p)^{-1} = \lambda_c(X_c'X_c + \lambda_c I_p)^{-1}$ we find the following equality with the notation $A_c = \lambda_c(X_c'X_c + \lambda_c I_p)^{-1}X_c'$

$$X'_{\mathcal{C}}M_{\mathcal{C},\Lambda_\ell} = \sum_c \lambda_c(X_c'X_c + \lambda_c I_p)^{-1}X_c' = \sum_c A_c.$$

Then we can rewrite $\hat\theta_g$ from (19) as

$$\hat\theta_g = \big(\frac{1}{C}\sum_c A_cX_c + \frac{\lambda_g}{C}I_p\big)^{-1}\big(\frac{1}{C}\sum_c A_cY_c\big) \tag{B.15}$$

$$= \big(\frac{1}{C}\sum_c A_cX_c + \frac{\lambda_g}{C}I_p\big)^{-1}\big(\frac{1}{C}\sum_c A_cX_c\big)\theta_g \tag{B.16}$$

$$+ \big(\frac{1}{C}\sum_c A_cX_c + \frac{\lambda_g}{C}I_p\big)^{-1}\big(\frac{1}{C}\sum_c A_cX_c\theta_{\ell,c}\big) \tag{B.17}$$

$$+ \big(\frac{1}{C}\sum_c A_cX_c + \frac{\lambda_g}{C}I_p\big)^{-1}\big(\frac{1}{C}\sum_c A_c\varepsilon_c\big). \tag{B.18}$$

That $\hat{\theta}_g \to \theta_g$ almost surely and in L^2 follows if we can show that the terms in (B.16)-(B.18) converges almost surely and in L^2 to θ_g, 0 and 0, respectively.

For the almost sure convergence, we first notice that $A_c X_c$ is positive definite with

$$A_c X_c \leq X_c' X_c$$

All entrances in a positive definite matrix B are numerically bounded by the largest eigenvalue of B, and in particular by the sum of all eigenvalues, which is equal to $\mathrm{tr}(B)$.

From this consideration we see that all entrances of $A_c X_c$ are numerically bounded by $\mathrm{tr}(X_c' X_c)$, which is integrable due to the integrability assumption on $(X_c' X_c)^2$. That gives integrability of $A_c X_c$. Similarly, $A_c X_c \theta_{\ell,c}$ and $A_c \varepsilon_c$ are integrable. To two latter expectations are both zero. Again using that $A_c X_c$ is positive definite, we have that $\mathbb{E}(X_c A_c)$ is positive definite, and in particular $(\mathbb{E}(A_c X_c))^{-1}$ exists. Now all three almost sure convergence statements follow from the strong law of large numbers.

For the L^2 convergence of the term in (B.16) we find, almost identically to the proof of Lemma 3, that

$$\begin{aligned}
&\mathbb{E}\Big(\big\|\big(\frac{1}{C}\sum_c A_c' X_c + \frac{\lambda_g}{C} I_p\big)^{-1}\big(\frac{1}{C}\sum_c A_c' X_c\big)\theta_g - \theta_g\big\|_2^2\Big) \\
&= \sigma_g^2 \mathbb{E}\Big(\mathrm{tr}\big(\frac{\lambda_g^2}{C^2}\big(\frac{1}{C}\sum_c A_c' X_c + \frac{\lambda_g}{C} I_p\big)^{-2}\big)\Big).
\end{aligned}$$

This is seen to converge to zero due to law of large numbers and dominated convergence, since

$$\frac{\lambda_g}{C}\big(\frac{1}{C}\sum_c A_c' X_c + \frac{\lambda_g}{C} I_p\big)^{-1} \leq I_p$$

The L^2-convergence of the terms in (B.17) and (B.18) is obtained similarly. Now the L^2-convergence of $\hat{\theta}_c^{\text{joint}}$ is obtained identically to the L^2-convergence of $\hat{\theta}_{c,\nu}^{\text{2stage}}$ in the proof of Lemma 3.

With arguments as in the proof of Proposition 1(ii)-(iii) we find that

$$
\begin{aligned}
&MSE(\hat{\theta}_c^{\text{joint}})\\
&\to MSE(\hat{\theta}_{c,\lambda}^{\text{local}}) + \nu^2\lambda^2\mathbb{E}\big(\|R_{c,\lambda}\theta_g\|_2^2\big) + 2\nu\lambda\mathbb{E}\big(\theta_g' R_{c,\lambda}(R_{c,\lambda}X_c'Y_c - \theta_c)\big)\\
&= MSE(\hat{\theta}_{c,\lambda_c}^{\text{local}}) + \nu(\nu-2)\sigma_g^2\mathbb{E}\big(\mathrm{tr}(\lambda_c^2 R_{c,\lambda}^2)\big),
\end{aligned}
$$

as $C \to \infty$, where the integrability of the limit follows from $\lambda_c R_{c,\lambda_c} \leq I_p$. This gives the result from (ii). □

C Bayesian Interpretations

C.1 Bayesian Interpretation of Generalized Elastic Net

The GENet has a Bayesian interpretation, similarly to a regular elastic net regression. To see this, assume for the time being that θ is drawn, independently of X, from the distribution with density $f(\theta)$, where

$$
f(\theta) \propto \exp(-\frac{1}{2}P_{\lambda,\alpha}(\theta - \nu\theta_0)).
$$

Thus the information that θ_0 is a good approximation to θ is included directly in the prior distribution. Furthermore, assume that

$$
f(Y \mid X, \theta) = N(Y \mid X\theta, I_N).
$$

Here $N(x \mid \xi, \Sigma)$ denotes the density of a Gaussian distribution with mean vector ξ and covariance matrix Σ evaluated in x. Consequently, the joint density of (Y, θ) given X is

$$
\begin{aligned}
f(Y, \theta \mid X) &\propto N(Y \mid X\theta, I_N) \cdot \exp(-\frac{1}{2}P_{\lambda,\alpha}(\theta - \nu\theta_0))\\
&\propto \exp\big(-\frac{1}{2}(Y - X\theta)'(Y - X\theta) - \frac{1}{2}P_{\lambda,\alpha}(\theta - \nu\theta_0)\big)
\end{aligned}
$$

In fact, this also determines the posterior density of θ given (Y, X) (when changing the above to the relevant function of θ, only the normalizing constant changes):

$$f(\theta \mid Y, X) \propto \exp\big(-\frac{1}{2}(Y - X\theta)'(Y - X\theta) - \frac{1}{2}P_{\lambda,\alpha}(\theta - \nu\theta_0)\big).$$

Maximizing this with respect to θ gives a MAP estimator for θ. Clearly, that is equivalent to minimizing

$$(Y - X\theta)'(Y - X\theta) + P_{\lambda,\alpha}(\theta - \nu\theta_0),$$

which gives the estimator in (10), so in fact,

$$\hat{\theta}^{\text{GENet}}_{c,\nu,\alpha,\lambda}(\theta_0) = \arg\max_{\theta \in \mathbb{R}^p} f(\theta \mid Y, X). \tag{C.1}$$

That is, $\hat{\theta}^{\text{GENet}}_{c,\nu,\alpha,\lambda}(\theta_0)$ is the most likely parameter in the posterior distribution given the observed data and prior knowledge about θ_0.

C.2 Bayesian Interpretation of Joint Model

Assume that $\theta_g, \theta_{\ell,1}, \ldots, \theta_{\ell,C}$ are independent with

$$f(\theta_g) = N(\theta_g \mid 0, \sigma_g^2 I_p)\,, \qquad f(\theta_{\ell,c}) = N(\theta_{\ell,c} \mid 0, \sigma_{\ell,c}^2 I_p),$$

and assume that given $X_{\mathcal{C}}$ and θ_g, θ_ℓ, the distribution of Y is Gaussian with

$$f(Y_{\mathcal{C}} \mid \theta_g, \theta_\ell, X_{\mathcal{C}}) = N(Y_{\mathcal{C}} \mid X_{\mathcal{C}}\theta_g + Z\theta_\ell, \sigma_\varepsilon^2 I_N).$$

Then the joint distribution of $(Y_{\mathcal{C}}, \theta_g, \theta_\ell)$ given $X_{\mathcal{C}}$ has density that satisfies

$$\begin{aligned}&f(Y_{\mathcal{C}}, \theta_g, \theta_\ell \mid X_{\mathcal{C}})\\ &\propto \exp\Big(-\frac{1}{2\sigma_\varepsilon^2}(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g - Z\theta_\ell)'(Y_{\mathcal{C}} - X_{\mathcal{C}}\theta_g - Z\theta_\ell) - \frac{1}{2\sigma_g^2}\theta_g'\theta_g - \sum_c \frac{1}{2\sigma_{\ell,c}^2}\theta_{\ell,c}'\theta_{\ell,c}\Big).\end{aligned}$$

Similarly to arguments in Section C.1 this is proportional to the posterior density of (θ_g, θ_ℓ) given $(Y_\mathcal{C}, X_\mathcal{C})$ as well, so

$$f(\theta_g, \theta_\ell \mid Y_\mathcal{C}, X_\mathcal{C}) \\ \propto \exp\Big(-\frac{1}{2\sigma_\varepsilon^2}(Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell)'(Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell) - \frac{1}{2\sigma_g^2}\theta_g'\theta_g - \sum_c \frac{1}{2\sigma_{\ell,c}^2}\theta_{\ell,c}'\theta_{\ell,c}\Big) \tag{C.2}$$

Clearly, maximizing this with respect to (θ_g, θ_ℓ) corresponds to minimizing

$$(Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell)'(Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell) + \frac{\sigma_\varepsilon^2}{\sigma_g^2}\theta_g'\theta_g + \sum_c \frac{\sigma_\varepsilon^2}{\sigma_{\ell,c}^2}\theta_{\ell,c}'\theta_{\ell,c}.$$

Setting $\lambda_g = \sigma_\varepsilon^2/\sigma_g^2$ and $\lambda_{\ell,c} = \sigma_\varepsilon^2/\sigma_{\ell,c}^2$ gives the minimization problem from (18).

C.3 Posterior Distribution for the Joint Model

We can describe the posterior distribution defined in equation (C.2) more explicitly. First, recalling that $\lambda_g = \sigma_\varepsilon^2/\sigma_g^2$ and $\lambda_{\ell,c} = \sigma_\varepsilon^2/\sigma_{\ell,c}^2$, we write

$$f(\theta_g, \theta_\ell \mid Y_\mathcal{C}, X_\mathcal{C}) \\ \propto \exp\Big(-\frac{1}{2\sigma_\varepsilon^2}\Big((Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell)'(Y_\mathcal{C} - X_\mathcal{C}\theta_g - Z\theta_\ell) - \lambda_g\theta_g'\theta_g - \theta_\ell'\Lambda_\ell\theta_\ell\Big)\Big) \\ \propto \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(\begin{pmatrix}\theta_g \\ \theta_\ell\end{pmatrix} - \begin{pmatrix}d_g \\ d_\ell\end{pmatrix}\right)' A \left(\begin{pmatrix}\theta_g \\ \theta_\ell\end{pmatrix} - \begin{pmatrix}d_g \\ d_\ell\end{pmatrix}\right)\right],$$

where

$$A = \begin{pmatrix} X_\mathcal{C}'X_\mathcal{C} + \lambda_g I_p & X_\mathcal{C}'Z \\ Z'X_\mathcal{C} & Z'Z + \Lambda_\ell \end{pmatrix}$$

and

$$\begin{pmatrix}d_g \\ d_\ell\end{pmatrix} = A^{-1}\begin{pmatrix}X_\mathcal{C}'Y_\mathcal{C} \\ Z'Y_\mathcal{C}\end{pmatrix}.$$

This shows that

$$\begin{pmatrix} \theta_g \\ \theta_\ell \end{pmatrix} \mid (Y_\mathcal{C}, X_\mathcal{C}) \sim N\left(\begin{pmatrix} d_g \\ d_\ell \end{pmatrix}, \sigma_\varepsilon^2 A^{-1}\right).$$

For applications, a training set could be used to estimate the parameter σ_ε^2 and determine the tuning parameters $\lambda_g, \lambda_{\ell,1}, \ldots, \lambda_{\ell,C}$.

D Dependence on Number of Training Observations

In this section we study if the number of training observations in different countries has an impact on model fits and performances. First, we examine to what extend the tuned transfer parameter ν, used in the soft transfer model and the 2stage model, depends on the number of training observations. The value of ν quantifies how much of the US (for soft transfer) or global (for 2stage) signal that should be used as a basis for the subsequent local fit. In Figure D.1 ν is tuned for each model and country using data up until 1999. It is seen that the magnitude of ν varies across countries, and not in a systematic way when the amount of training data increases. In most countries, the models result in rather similar values of ν. One notable exception is Japan, where the 2stage value of ν is twice as big as the soft transfer value. Since Japan has the second largest amount of training data, the Japanese data also plays a substantial role in the global model fit that is used as a basis for the 2stage transfer model. Consequently, much more of the global signal is transferred in comparison with the US signal used in the soft transfer model.

In Table D.1 we regress the performance of the local model in each country, as measured by R^2_{OOS} and the Sharpe ratio of the resulting long-short portfolio, on the number of training observations. The regression is both with and without a dummy variable indicating if the country is among the 7 countries with the smallest amount of training data. Somewhat surprisingly, there is no significant effect of the number of observations available when using a local model. This does not mean that more data in the individual country could not have improved the performance. It is merely a result of a large variation in ability to predict across countries. We have similarly regressed the gap in performance between the hard and

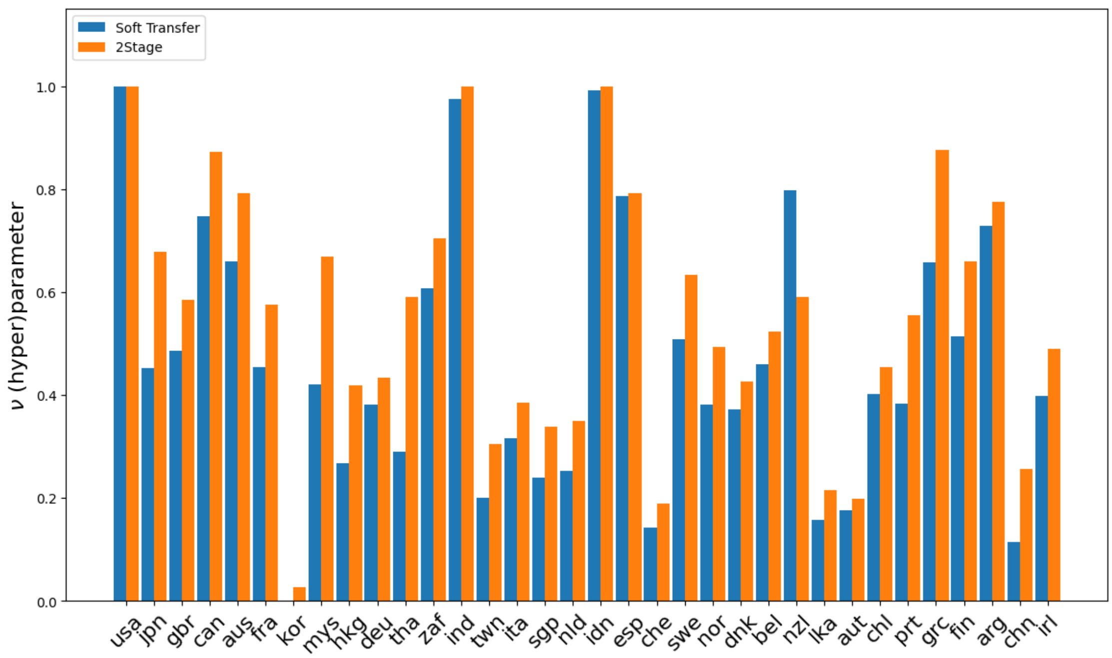

Figure D.1: **Tuned Hyperparameter ν across Countries.** This figure plots, for each country, the tuned hyperparameter ν for the soft transfer model and the 2stage model, using data up until 1999. Countries are sorted by the amount of available training data

	Local R^2_{OOS}		Hard - Local R^2_{OOS}		Local Sharpe		Hard - Local Sharpe		$\mathcal{G}_c$		ν Soft		ν 2Stage	
Intercept	0.06	0.07	0.09	0.09	0.31^{***}	0.32^{***}	0.26^{**}	0.26^{**}	0.96^{***}	0.95^{***}	0.46^{***}	0.46^{***}	0.59^{***}	0.58^{***}
	(1.14)	(1.23)	(0.78)	(0.79)	(5.12)	(4.56)	(2.48)	(2.34)	(131.07)	(120.77)	(7.21)	(6.98)	(9.48)	(9.19)
# Obs (Std.)	0.03	0.02	-0.52	-0.55	0.00	0.00	-0.80^{*}	-0.73	-0.01	-0.01	0.09	0.11	0.26	0.32
	(0.50)	(0.43)	(-1.09)	(-1.09)	(0.03)	(0.01)	(-1.75)	(-1.50)	(-1.39)	(-1.18)	(0.33)	(0.38)	(0.97)	(1.14)
Bottom 7		-0.06		-0.04		-0.03		0.11		0.03^{*}		0.03		0.08
		(-0.49)		(-0.19)		(-0.16)		(0.58)		(1.96)		(0.23)		(0.79)
Observations	34	34	33	33	34	34	33	33	34	34	33	33	33	33
R^2	0.01	0.02	0.04	0.04	0.00	0.00	0.09	0.10	0.06	0.16	0.00	0.01	0.03	0.05

Table D.1: **Effect of Number of Training Observations on Results**. The performance of the local model (measured in R^2_{OOS} and Sharpe ratio of the resulting long-short portfolio) is regressed on the number of observations in the training data, standardized across countries. The table includes a specification with a dummy variable indicating whether the country is in the bottom 7 of the number of training observations. Similarly, the gap in performance between the hard transfer model and the local model, the globalness score $\mathcal{G}_c$, and the tuned hyperparameter ν from the soft transfer model and two-stage model are regressed on the standardized number of observations.

the local model on the number of training observations. Though the hard model generally performs better, there is no additional effect of the amount of available data.

In the regression of the country-specific globalness score, we see that the 7 countries with the least amount of training data appear to be more global. This is most likely a result of the local amount of training data being so little that the local estimation vanishes and only the global parameters are used. Finally, we see that the tuned parameter ν depends positively on the amount of training data in the 2stage model, whereas this effect is not significant for the soft transfer model. This difference in effect is similar to the difference in ν-parameters for Japan observed in Figure D.1: The countries with much training data play an important role in estimating the global parameters. Consequently, the transfer parameter is larger when transferring from the global estimate.

In Figure D.2, we depict the global number of observations in the rolling subset of test data used to produce Figure 7. Clearly, the number of observations increases monotonically, so the small fluctuations seen in Figure 7 cannot be attributed solely to the number of observations.

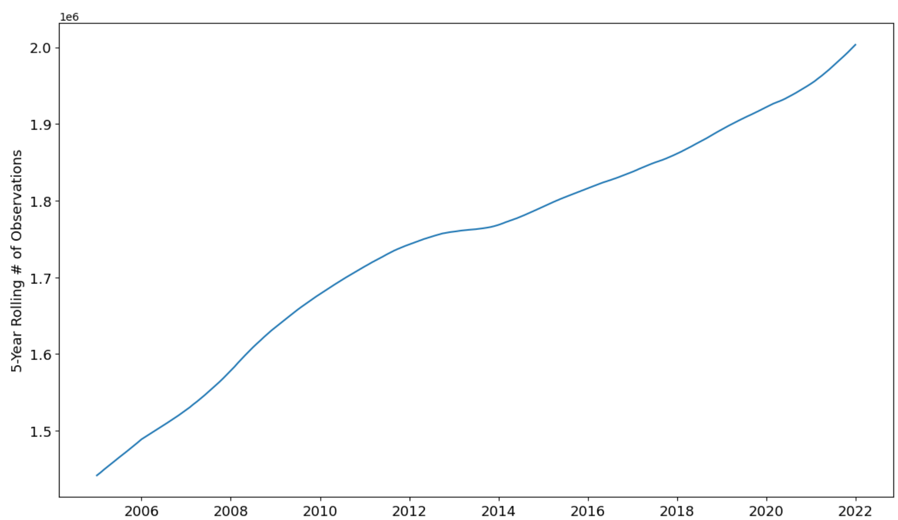

Figure D.2: **5-Year Rolling Number of Observations** This figure plots the 5-year rolling number of observations, globally.

E Hyperparameter Spaces

Table E.1 shows the hyperparameter space that the Bayesian optimization searches over. All L^2 penalization parameters have the same space, while the penalty parameter for elastic net models has a slightly different space. The reason for this discrepancy is that higher values of λ result in all coefficients being exactly 0 with even a modest L^1 ratio. Since the scaling is different for RFF features, their ridge penalty also has a slightly different search space.

Linear Models					
First-Stage Models		Transfer Models		Joint Model	
Ridge penalty, λ	$10^{[-3,5]}$	Penalty, λ	$10^{[-3,0]}$	Global penalty, λ_g	$10^{[-3,5]}$
		L^1 ratio	$[0,1]$	Local penalty, $\lambda_{c,\ell}$	$10^{[-3,5]}$
		Transfer parameter, ν	$[0,1]$		
Non-Linear Models					
XGB		RFF			
colsample_bytree	$[1/111, 1]$	Ridge penalty, λ	$10^{[-5,10]}$		
max_depth	$\{1, 2, ..., 7\}$	Feature Vol, η	$e^{[-3,0]}$		
subsample	$[0.2, 1]$				
L^2 regularization, λ	$10^{[-2,2]}$				
Learning rate, η	$[0.01, 0.15]$				

Table E.1: **Hyperparameter Space**. This table shows the hyperparameter spaces we tune over. We use Bayesian optimization for tuning, using the Optuna package (Akiba et al., 2019).

F Characteristics

Table F.1 shows all characteristics used in the empirics, and their categorization. The classification is based on that of Chen et al. (2022) for most of the characteristics. For those that are not included in Chen et al. (2022), we seek to do the classification using their methodology. Finally, we cross-check their classifications, reclassifying `betabab_1260d` and `ami_126d` as risk-based characteristics rather than mispricing.

Mispricing Inv	Mispricing Fund	Mispricing Misc	Risk	Agnostic
at_gr1	age	at_turnover	aliq_at	at_be
be_gr1a	dbnetis_at	debt_me	aliq_mat	at_me
capx_gr1	fnl_gr1a	dolvol_126d	ami_126d	be_me
coa_gr1a	ivol_capm_21d	ebit_bev	beta_60m	bev_mev
col_gr1a	ivol_capm_252d	ebit_sale	beta_dimson_21d	chcsho_12m
cowc_gr1a	lti_gr1a	ebitda_mev	betabab_1260d	cop_at
inv_gr1a	ncol_gr1a	fcf_me	betadown_252d	cop_atl1
lnoa_gr1a	nfna_gr1a	mispricing_perf	bidaskhl_21d	dolvol_var_126d
mispricing_mgmt	noa_at	ni_be	cash_at	eqnpo_12m
ncoa_gr1a	pi_nix	ni_me	corr_1260d	gp_at
nncoa_gr1a	prc_highprc_252d	o_score	coskew_21d	gp_atl1
noa_gr1a	ret_12_1	ocf_at	div12m_me	iskew_capm_21d
oaccruals_at	ret_3_1	ocf_at_chg1	emp_gr1	iskew_ff3_21d
oaccruals_ni	ret_6_1	ocf_me	eq_dur	ival_me
ret_60_12	ret_9_1	qmj_prof	inv_gr1	ivol_ff3_21d
sale_gr1	rmax1_21d	qmj_safety	opex_at	kz_index
taccruals_at	rmax5_21d	sale_bev	ppeinv_gr1a	market_equity
taccruals_ni	rmax5_rvol_21d	sale_emp_gr1	prc	netdebt_me
	turnover_126d	tax_gr1a	tangibility	op_at
			zero_trades_126d	op_atl1
			zero_trades_21d	ope_be
			zero_trades_252d	ope_bel1
				ret_1_0
				ret_12_7
				rskew_21d
				rvol_21d
				rvol_252d
				sale_me
				seas_1_1an
				seas_1_1na
				seas_2_5an
				seas_2_5na
				turnover_var_126d

Table F.1: This table shows the classification of the 111 characteristics into the subgroups Mispricing Inv, Mispricing Fund, Mispricing Misc, Risk and Agnostic